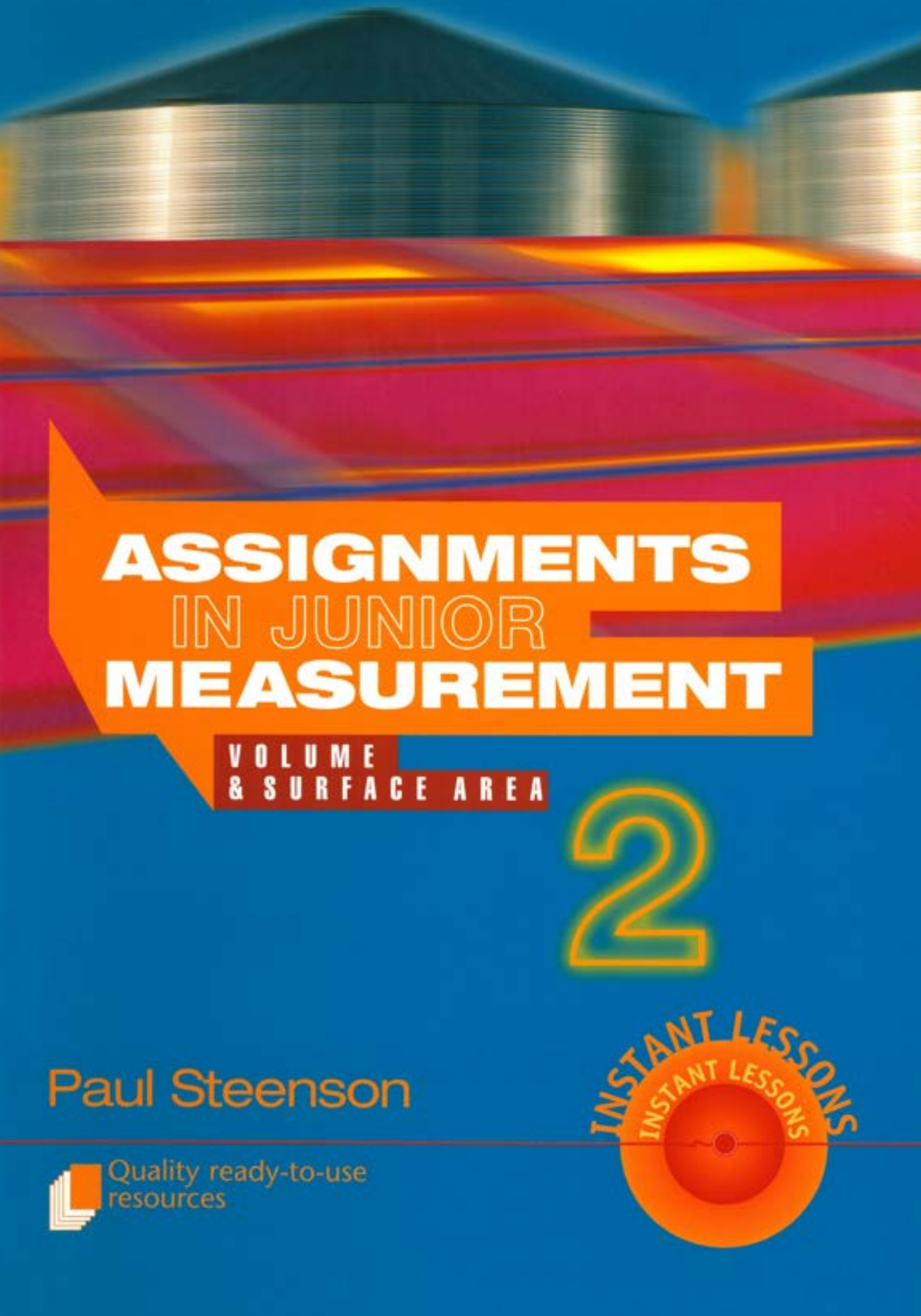




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**VOLUME
& SURFACE AREA**

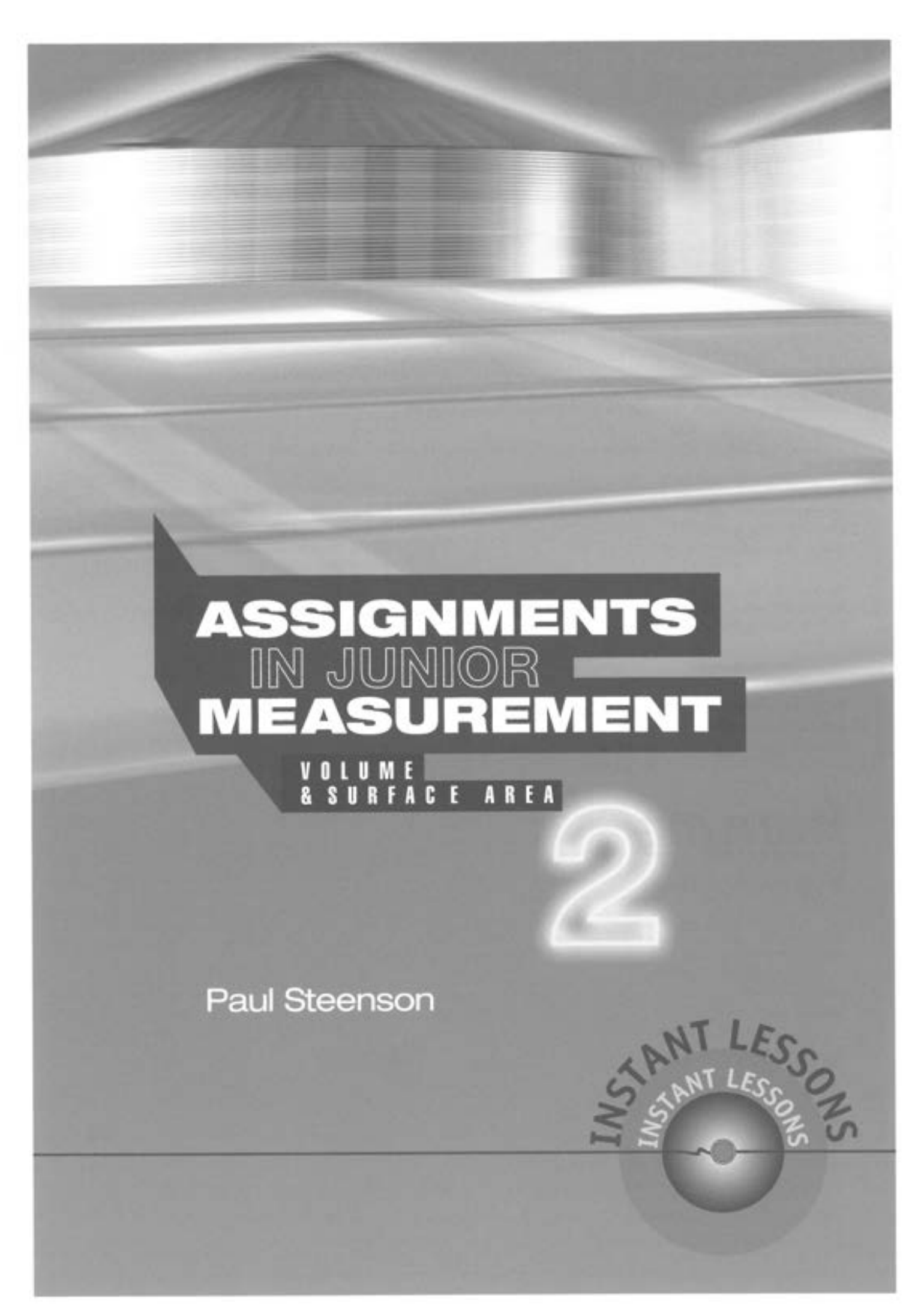
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ASSIGNMENTS IN JUNIOR **MEASUREMENT**

VOLUME
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Acknowledgements

My thanks to Kamal, Nathan, Darren and Chris for working through the draft worksheets. Hopefully we have removed all the bugs!

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Introduction

This book provides a systematic and comprehensive coverage of selected topics from the junior mathematics curriculum. It is independent of any textbook and is not linked to any syllabus.

Each worksheet follows a format comprising three sections:

Section A consists of a set of introductory questions or review of previous knowledge which can be used as a lesson starter or 'warm up' exercise.

Section B provides skills development or drill with graded questions in the focus area of the worksheet.

Section C provides further practice in the form of different types of questions, a riddle, challenge or extension questions.

Review worksheets are also included in each topic area.

There are minimal examples or explanations as teachers have different methods or working that they would like their students to follow.

Most worksheets could be used by students in a range of years or courses. The teacher may decide to use only one or two sections from a worksheet in a lesson or as a homework assignment. The worksheets are designed to be flexible to meet the needs of the teacher and students.

A time required for students to complete each worksheet is not suggested as this will vary significantly depending on the year level of the students and their level of mathematical skill. The teacher is the best person to make this judgement.

Contents

1	Volume of a right prism	4
2	Volume of a rectangular prism	6
3	Volume of a triangular prism	8
4	Volume of a cylinder	10
5	Review: volumes of prisms	12
6	Volume of a pyramid	14
7	Volume of a cone	16
8	Volume of a sphere	18
9	Review: volume	20
10	Surface area of a rectangular prism	23
11	Surface area of triangular prism	25
12	Surface areas of various prisms	27
13	Surface area of a cylinder	30
14	Surface area of a pyramid	33
15	Surface area of a cone	35
16	Surface area of a sphere	37
17	Review: surface areas of 3D shapes	39
18	Review: volumes and surface areas	41
19	Equivalent units	44
	Answers	45
	Formulae	48

Volume of a right prism

1.1

A

Prisms have a uniform cross-section. This means that a slicing machine would cut slices that are all the same shape and size. In a right prism, the other faces of the solid are at right angles to the cross-section. (The cross-section is shown by the shaded face).

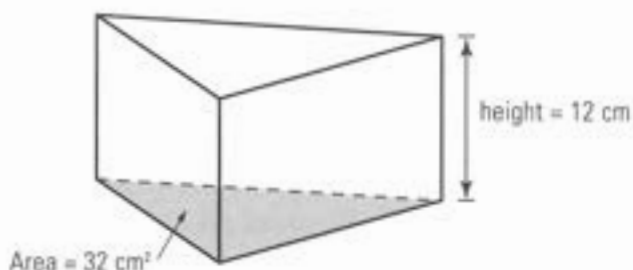
To calculate the volume of a right prism:

$$\text{Volume} = \text{Area} \times \text{perpendicular height}$$

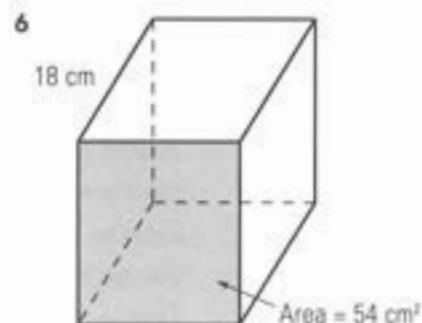
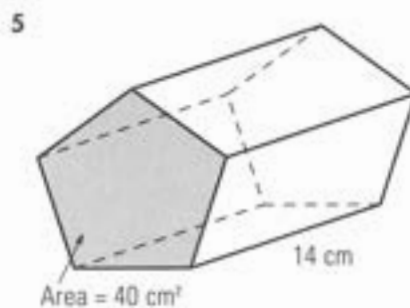
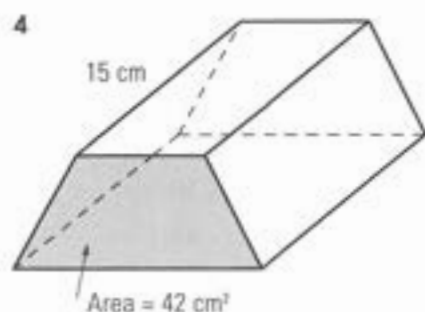
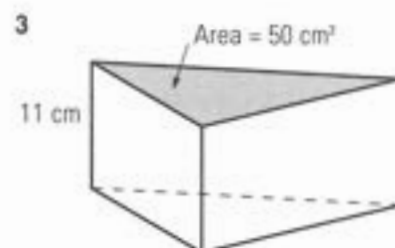
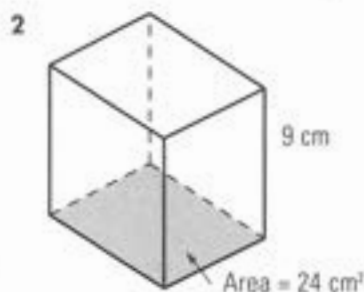
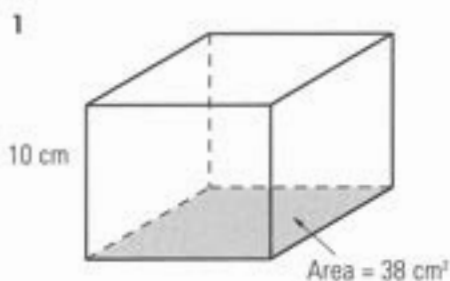
OR $V = A \times h$

EXAMPLE: The volume of this triangular prism

$$\begin{aligned} V &= 32 \times 12 \\ &= 384 \text{ cm}^3 \end{aligned}$$

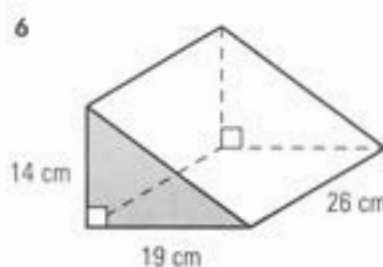
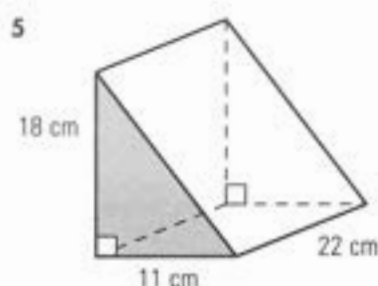
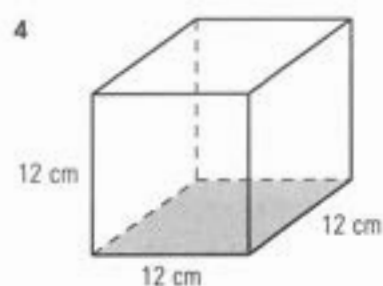
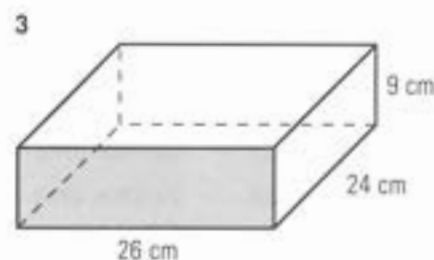
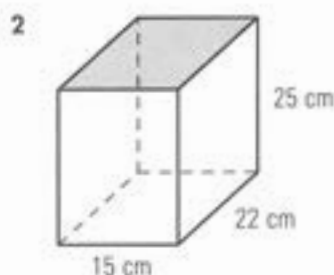
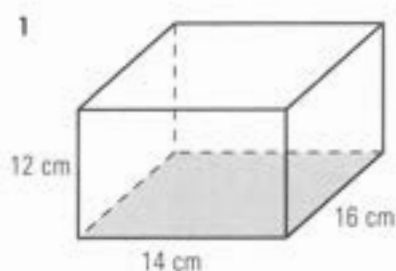


Use the formula $V = A \times h$ to calculate the volume of each prism.

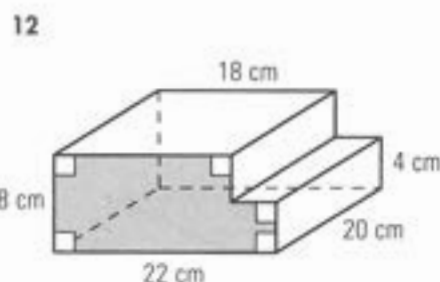
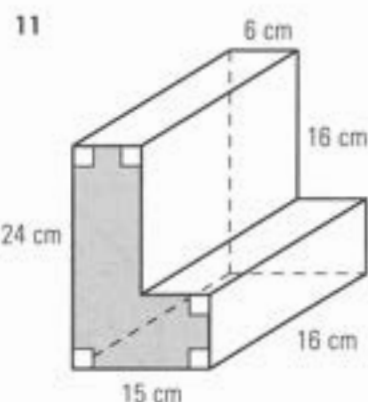
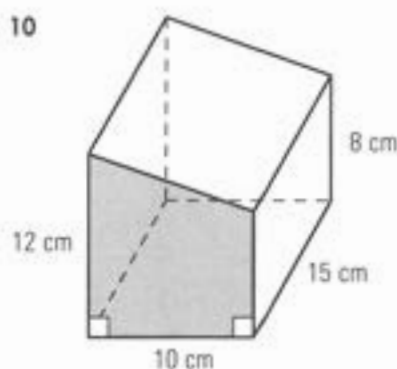
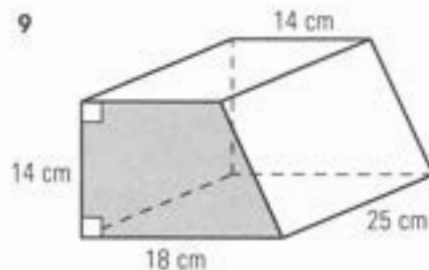
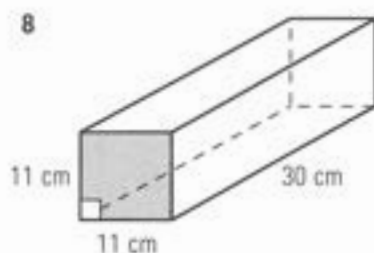
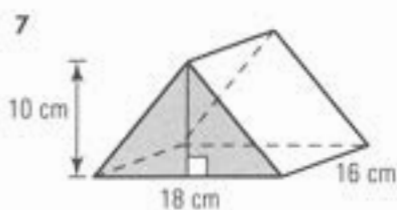


B

Use the formula $V = A \times h$ to calculate the volume of each prism.



Use the formula $V = A \times h$ to calculate the volume of each prism.



C For each of the following, V is the volume of a right prism and A is the area of its cross-section. Calculate the perpendicular height of each prism.

1 $V = 1080 \text{ cm}^3$ $A = 90 \text{ cm}^2$

2 $V = 935 \text{ cm}^3$ $A = 85 \text{ cm}^2$

3 $V = 1650 \text{ cm}^3$ $A = 110 \text{ cm}^2$

4 $V = 1440 \text{ cm}^3$ $A = 160 \text{ cm}^2$

5 $V = 1504 \text{ cm}^3$ $A = 94 \text{ cm}^2$

6 $V = 1610 \text{ cm}^3$ $A = 115 \text{ cm}^2$

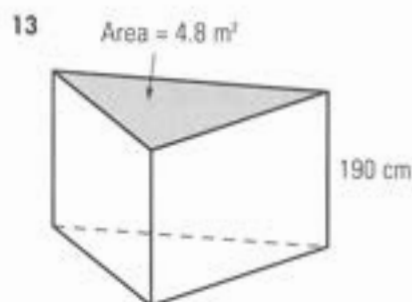
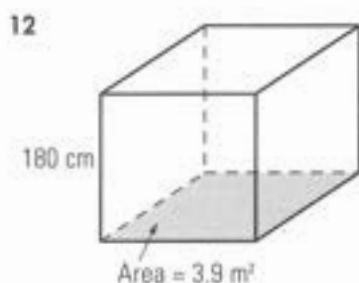
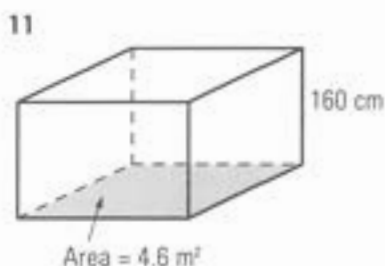
7 $V = 2880 \text{ cm}^3$ $A = 240 \text{ cm}^2$

8 $V = 15.3 \text{ m}^3$ $A = 8.5 \text{ m}^2$

9 $V = 44.5 \text{ m}^3$ $A = 17.8 \text{ m}^2$

10 $V = 23.04 \text{ m}^3$ $A = 7.2 \text{ m}^2$

Calculate the volume of each prism in **cubic metres**.
(Hint: change all measurements to metres **before** you multiply.)



For each of the following, A is the area of the cross-section of a right prism and h is its perpendicular height. Calculate the volume of each prism.

14 $A = 5.3 \text{ m}^2$ $h = 210 \text{ cm}$

15 $A = 9.5 \text{ m}^2$ $h = 350 \text{ cm}$

16 $A = 6 \text{ m}^2$ $h = 255 \text{ cm}$

17 $A = 3 \text{ m}^2$ $h = 185 \text{ cm}$

18 $A = 4.9 \text{ m}^2$ $h = 265 \text{ cm}$

19 $A = 7.2 \text{ m}^2$ $h = 305 \text{ cm}$

Volume of a rectangular prism

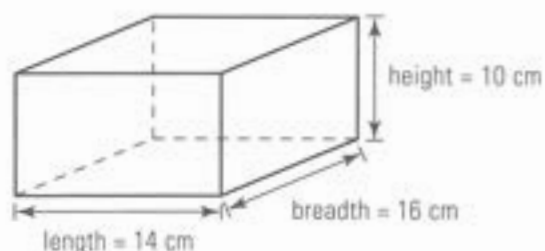
2.1

The cross-section of a rectangular prism is a rectangle. All of the faces are rectangles and opposite pairs of faces are exactly the same.

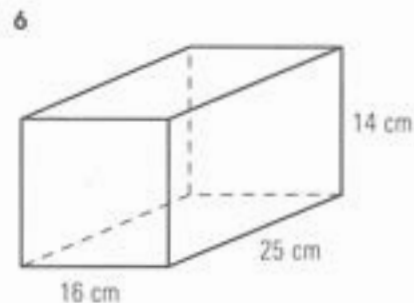
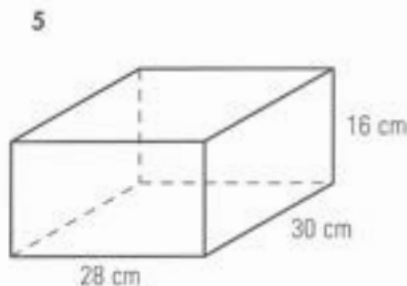
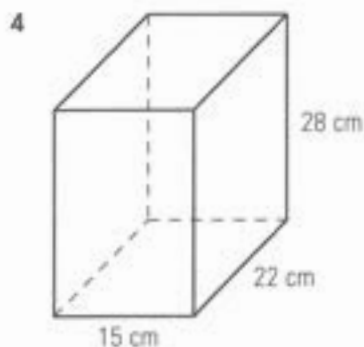
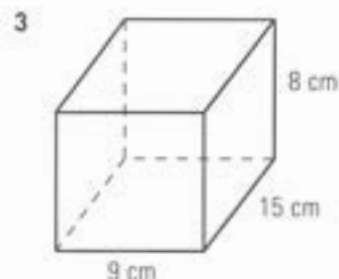
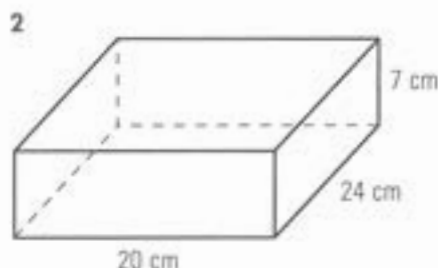
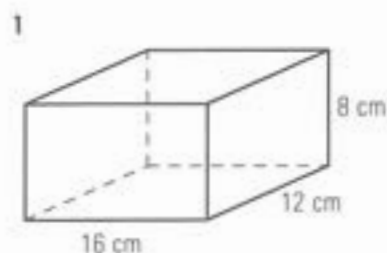
The volume of a rectangular prism is calculated using the formula:

$$\text{Volume} = \text{length} \times \text{breadth} \times \text{height}$$

EXAMPLE: $V = 14 \times 16 \times 10$
 $= 2240 \text{ cm}^3$



A Calculate the volume of each rectangular prism.



B Use the given dimensions to calculate the volume of each rectangular prism.

1 length = 18 cm
breadth = 14 cm
height = 12 cm

4 length = 26 cm
breadth = 12 cm
height = 15 cm

7 length = 20 cm
breadth = 15 cm
height = 22 cm

10 length = 55 mm
breadth = 40 mm
height = 35 mm

13 length = 3.5 m
breadth = 2.5 m
height = 1.8 m

2 length = 15 cm
breadth = 22 cm
height = 9 cm

5 length = 16 cm
breadth = 14 cm
height = 11 cm

8 length = 24 cm
breadth = 18 cm
height = 7 cm

11 length = 80 mm
breadth = 65 mm
height = 25 mm

14 length = 4.8 m
breadth = 1.6 m
height = 2.3 m

3 length = 14 cm
breadth = 11 cm
height = 8 cm

6 length = 22 cm
breadth = 16 cm
height = 13 cm

9 length = 85 mm
breadth = 50 mm
height = 30 mm

12 length = 170 mm
breadth = 55 mm
height = 85 mm

15 length = 2.6 m
breadth = 2.2 m
height = 3.1 m

Use the given dimensions to calculate the volume of each rectangular prism.

16 length = 3.2 m
breadth = 1.9 m
height = 2.7 m

17 length = 4.5 m
breadth = 3 m
height = 1.2 m

18 length = 5.2 m
breadth = 1.8 m
height = 2.5 m

19 length = 2.8 m
breadth = 1.5 m
height = 1.4 m

20 length = 3.9 m
breadth = 2.3 m
height = 2.7 m

C Calculate the unknown dimension for each rectangular prism.

1 Volume = 720 cm^3 , length = 15 cm,
height = 6 cm, breadth = ?

2 Volume = 1080 cm^3 , length = 12 cm,
breadth = 10 cm, height = ?

3 Volume = 4800 cm^3 , breadth = 20 cm,
height = 10 cm, length = ?

4 Volume = 9000 cm^3 , length = 15 cm,
height = 20 cm, breadth = ?

5 Volume = 2400 cm^3 , breadth = 12 cm,
height = 8 cm, length = ?

6 Volume = 5376 cm^3 , length = 28 cm,
breadth = 16 cm, height = ?

7 Volume = 14.4 m^3 , breadth = 2.4 m,
height = 1.5 m, length = ?

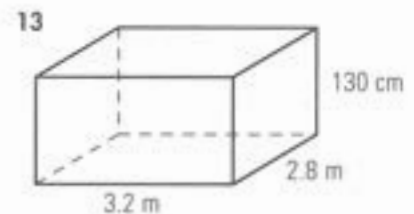
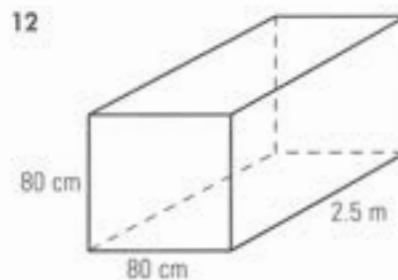
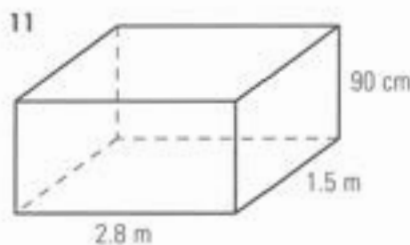
8 Volume = 11.44 m^3 , length = 2.6 m,
height = 2.2 m, breadth = ?

9 Volume = 5.61 m^3 , length = 2.2 m,
breadth = 1.7 m, height = ?

10 Volume = 7.904 m^3 , length = 3.8 m,
height = 1.3 m, breadth = ?

Calculate the volume of each prism in **cubic metres**.

(Hint: change all measurements to metres **before** you multiply.)



14 length = 4.2 m
breadth = 1.8 m
height = 195 cm

15 length = 4.5 m
breadth = 2.6 m
height = 240 cm

16 length = 3.6 m
breadth = 190 cm
height = 2.2 m

17 length = 5.2 m
breadth = 250 cm
height = 3.4 m

18 length = 4 m
breadth = 230 cm
height = 180 cm

19 length = 4.8 m
breadth = 350 cm
height = 250 cm

20 length = 160 cm
breadth = 130 cm
height = 155 cm

21 length = 210 cm
breadth = 180 cm
height = 85 cm

22 length = 260 cm
breadth = 140 cm
height = 230 cm

Volume of a triangular prism

3.1

A

The cross-sections and the two end-faces of a **triangular prism** are triangles, the same shape and the same size. All other faces are rectangles.

To calculate the volume of a triangular prism:

$$\text{Volume} = \text{Area} \times \text{height}$$

OR

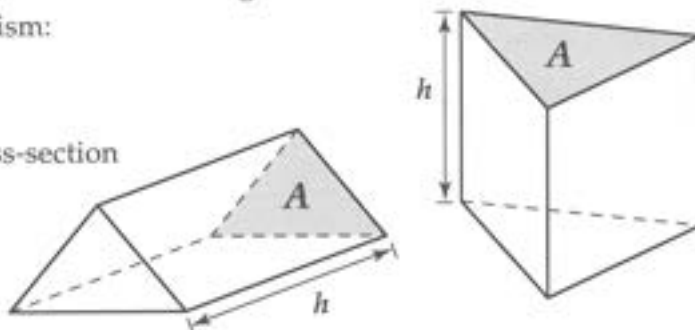
$$V = A \times h$$

where

A = area of the triangular cross-section

and

h = the perpendicular height (or distance) between the triangular end-faces.

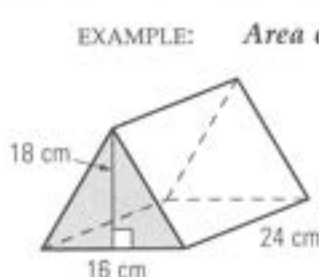


Calculate the volume of each triangular prism.

- | | |
|--|--|
| 1 triangular cross-section = 24 cm^2 , height = 9 cm | 2 triangular cross-section = 56 cm^2 , height = 8 cm |
| 3 triangular cross-section = 82 cm^2 , height = 15 cm | 4 triangular cross-section = 136 cm^2 , height = 20 cm |
| 5 triangular cross-section = 150 cm^2 , height = 24 cm | 6 triangular cross-section = 145 cm^2 , height = 22 cm |

B

If you are given dimensions of the triangular cross-section, first calculate the area of the triangle then calculate the volume.



EXAMPLE: $\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$

$$= \frac{1}{2} \times 16 \times 18$$

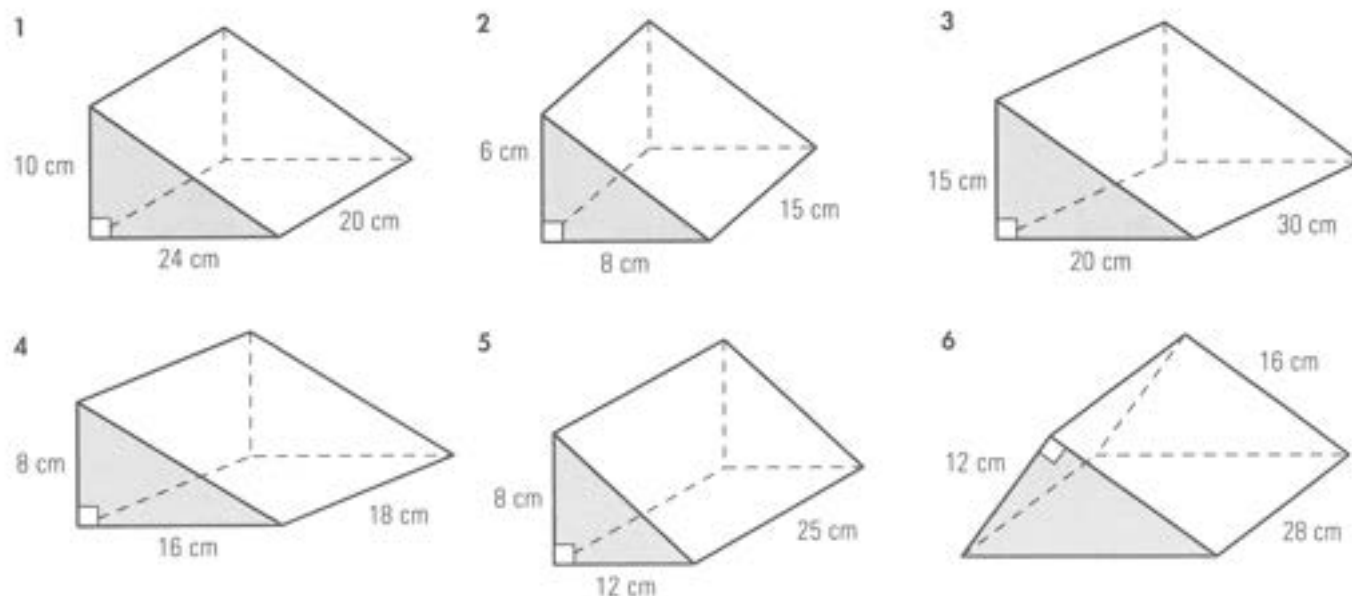
$$= 144 \text{ cm}^2$$

$\text{Volume} = A \times h$

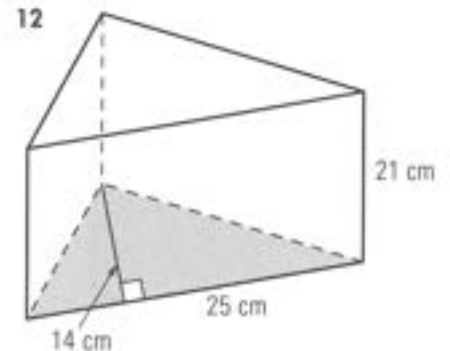
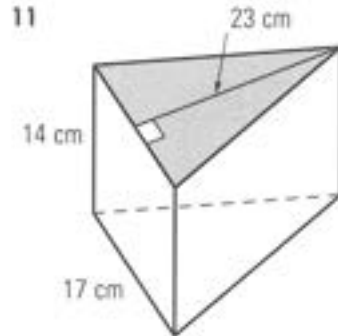
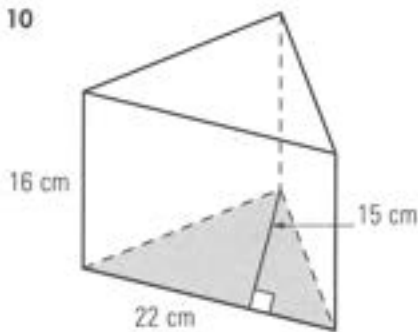
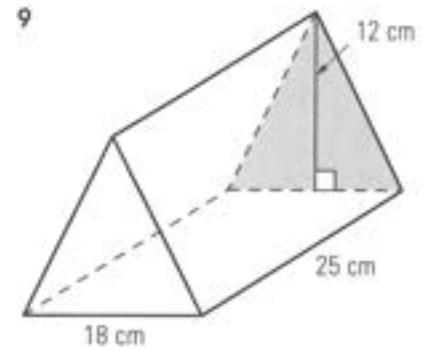
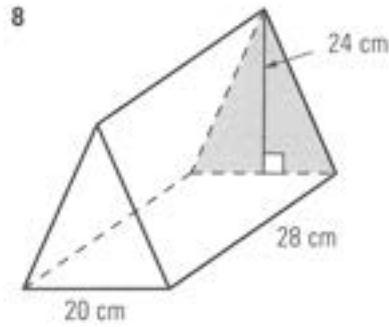
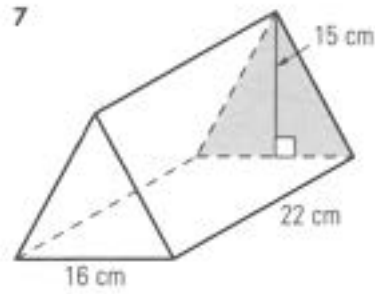
$$= 144 \times 24$$

$$= 3456 \text{ cm}^3$$

Calculate the volume of each triangular prism.
(Hint: first calculate the area of the triangular cross-section.)



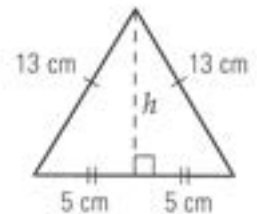
Calculate the volume of each triangular prism. (Hint: first calculate the area of the cross-section.)



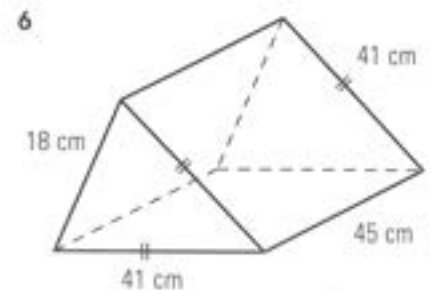
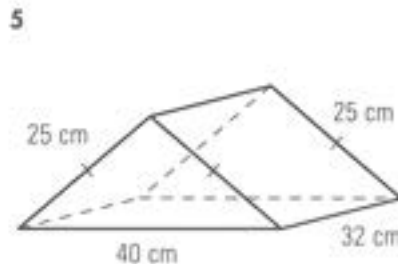
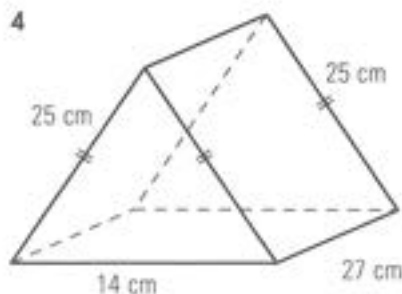
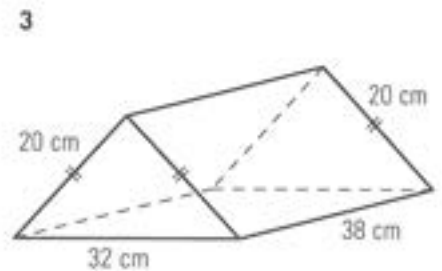
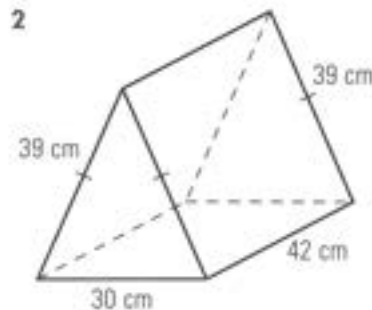
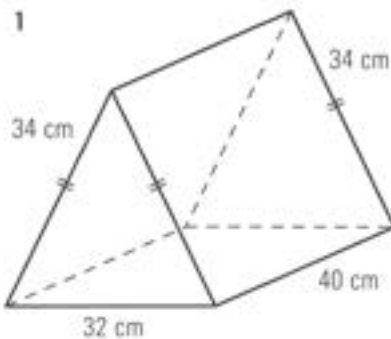
C In an isosceles triangle, the axis of symmetry can be used to calculate the height of the triangle. The axis of symmetry forms two right-angled triangles. By applying **Pythagoras' theorem**, you can calculate the height.

EXAMPLE:

$$\begin{aligned} h^2 &= 13^2 - 5^2 \\ &= 144 \\ h &= \sqrt{144} \\ &= 12 \text{ cm} \end{aligned}$$



The cross-sections of these prisms are isosceles triangles. Calculate the volume of each prism.



Volume of a cylinder

4.1

A

A cylinder is a prism with a circular cross-section.
The volume of a cylinder is calculated using the formula:

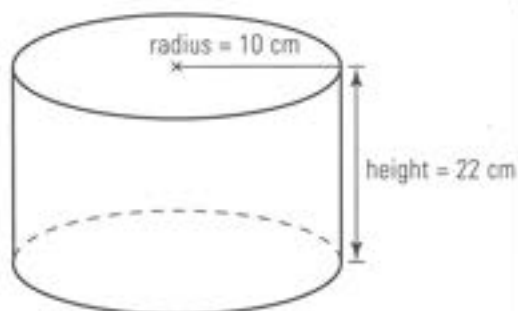
$$\text{Volume} = \pi \times (\text{radius})^2 \times \text{height}$$

OR $V = \pi r^2 h$

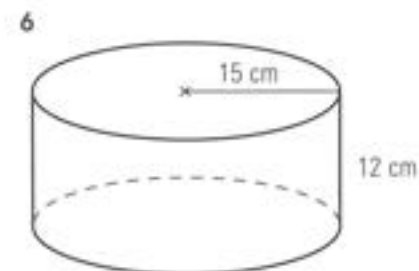
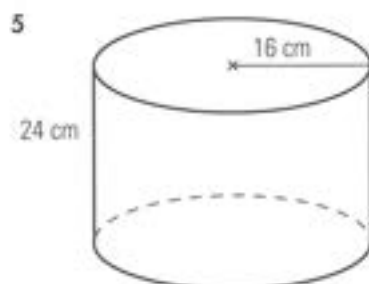
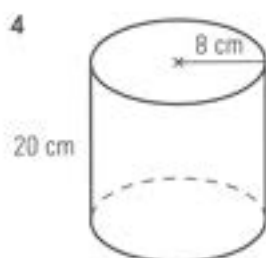
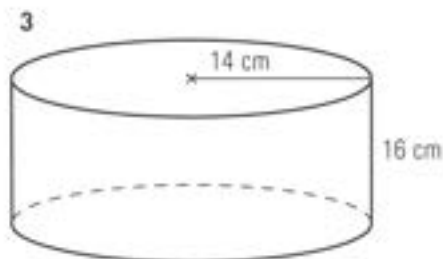
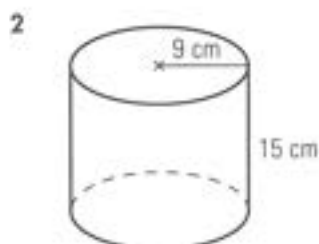
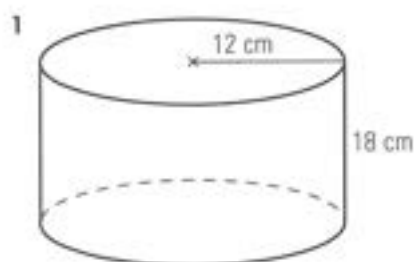
EXAMPLE: $V = \pi \times 10^2 \times 22$

$$= 6\,912 \text{ cm}^3$$

(to the nearest cubic centimetre)

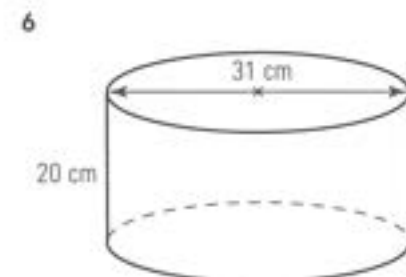
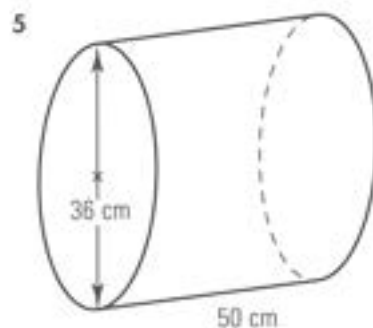
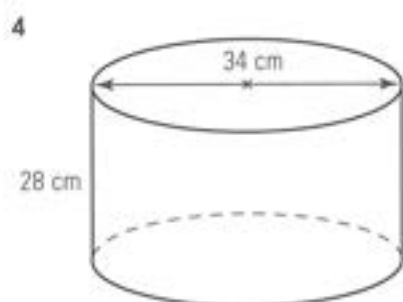
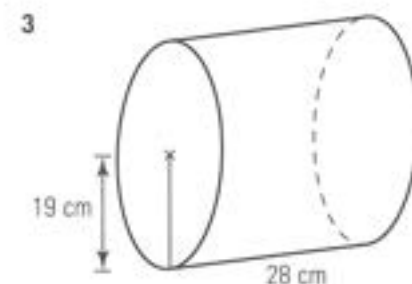
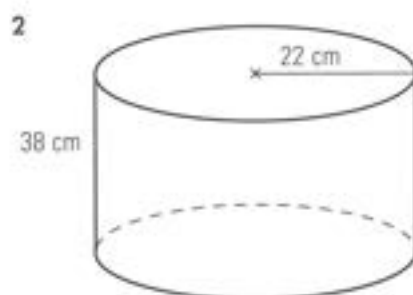
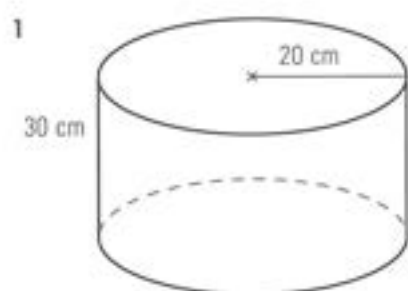


Calculate the volume of each cylinder to the nearest cubic centimetre.

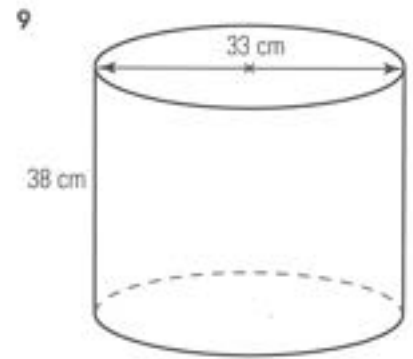
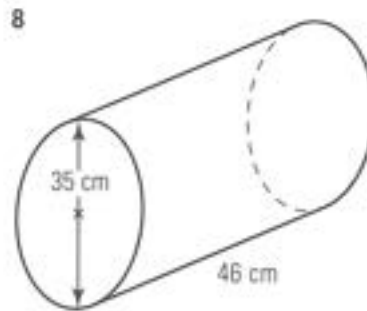
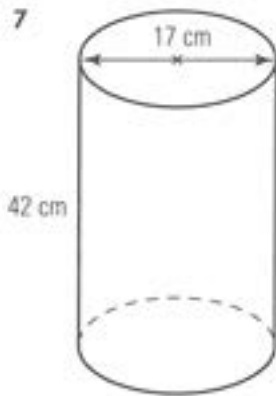


B

Calculate the volume of each cylinder, correct to one decimal place.



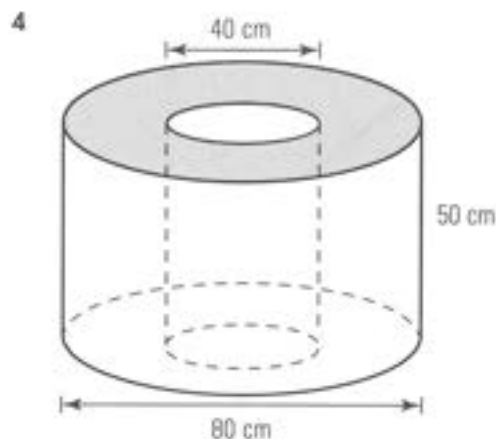
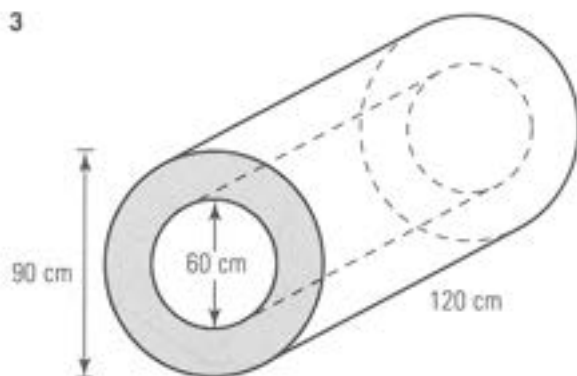
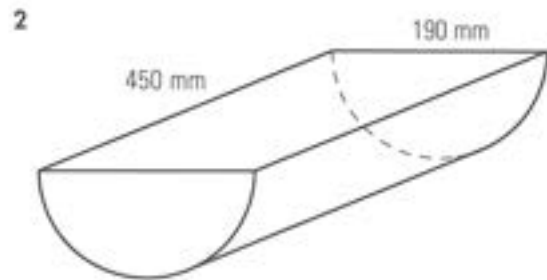
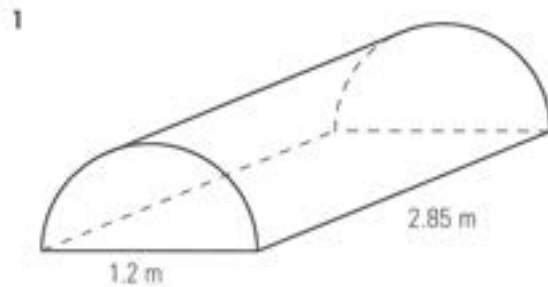
Calculate the volume of each cylinder, correct to one decimal place.



- 10 cylinder: radius 1.9 m, height 2.7 m
 12 cylinder: radius 85 mm, height 200 mm
 14 cylinder: diameter 2.4 m, height 3.6 m
 16 cylinder: diameter 105 mm, height 260 mm

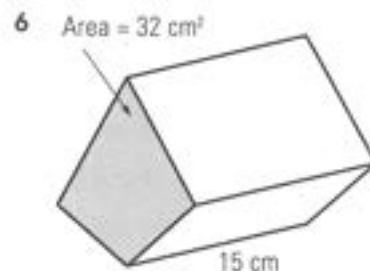
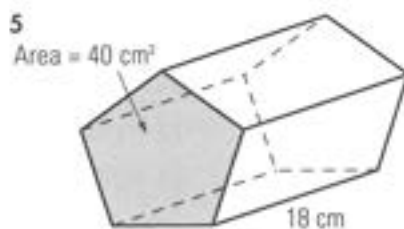
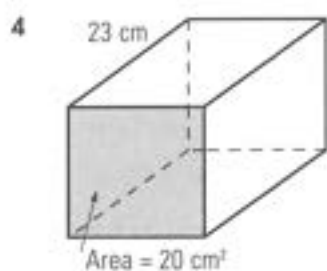
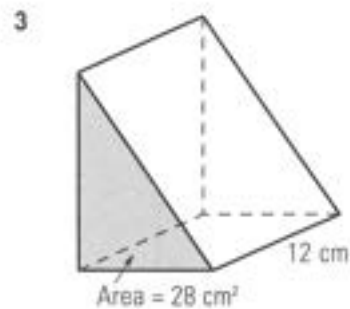
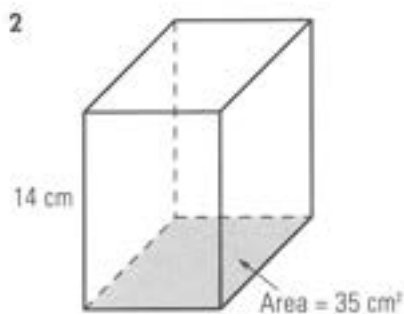
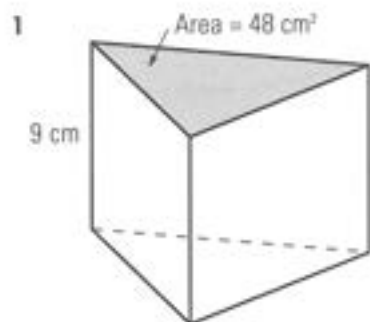
- 11 cylinder: radius 0.8 m, height 1.35 m
 13 cylinder: radius 65 mm, height 140 mm
 15 cylinder: diameter 1.6 m, height 2.95 m
 17 cylinder: diameter 90 mm, height 375 mm

C Calculate the volume of each solid, correct to one decimal place.

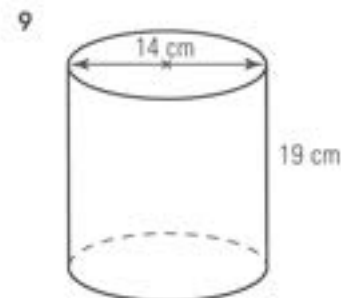
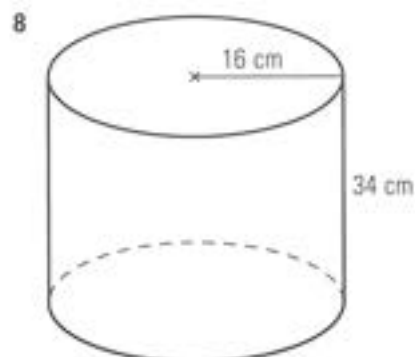
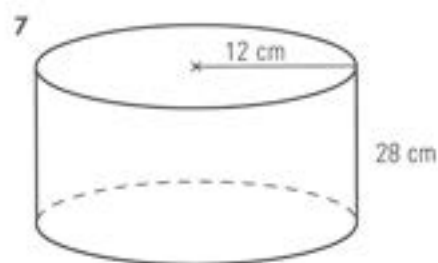
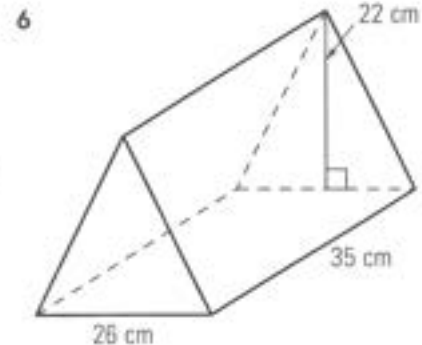
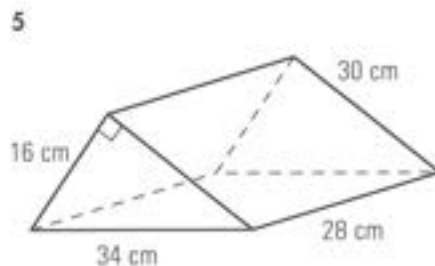
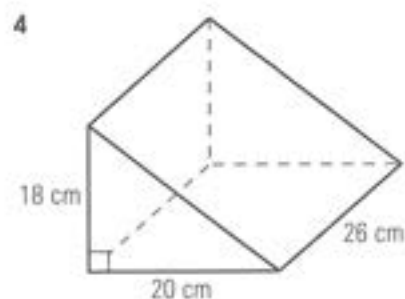
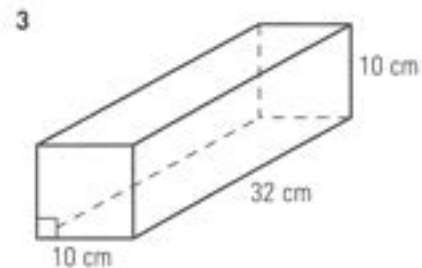
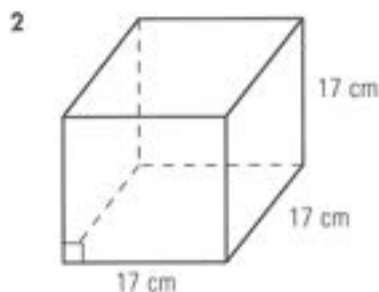
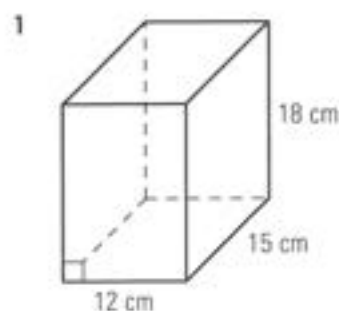


- 5 A cylinder has a volume of 509 cm^3 and a diameter of 9 cm. Calculate its height to the nearest centimetre.
- 7 A cylindrical storage tank has a capacity of 6000 litres. The diameter of the tank is 2 metres. How deep will the water be when the tank is half full?
- 6 A cylindrical can is 10 cm tall and its diameter is 10 cm.
 a Calculate the volume of the can.
 b Does the can hold more than 1 litre of water? (1 litre = 1000 cm^3)
- 8 A cylindrical container has a diameter of 8 cm. If 800 mL of water is poured into the container, how deep will the water be?

A Use the formula $V = A \times h$ to calculate the volume of each prism.



B Calculate the volume of each prism (correct to one decimal place, where necessary.)



Calculate the volume of each prism (correct to one decimal place, where necessary.)

10 rectangular prism: edges 28 cm, 24 cm, 19 cm

12 square prism: edges 16 cm, 16 cm, 25 cm

14 cylinder: radius 11 cm, height 18 cm

16 rectangular prism: edges 2.6 m, 1.4 m, 1.2 m

18 cube: edges 1.9 m

20 cylinder: diameter 1.5 m, height 1.8 m

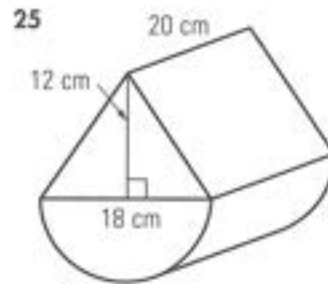
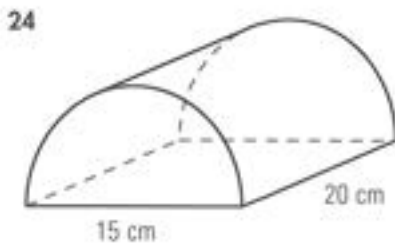
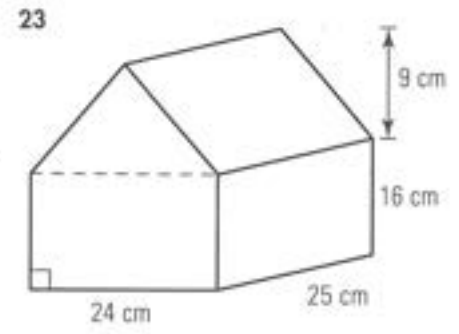
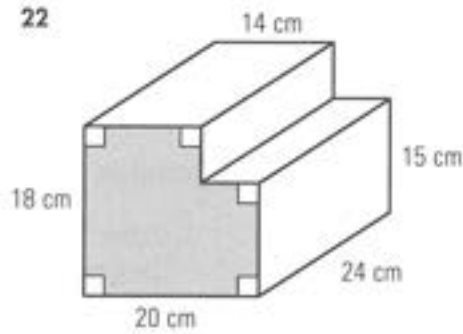
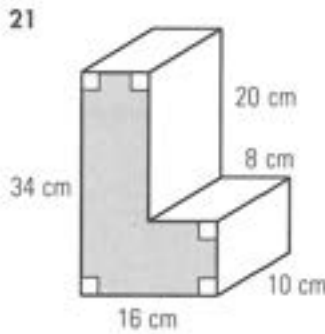
11 cube: edges 13 cm

13 rectangular prism: edges 32 cm, 21 cm, 16 cm

15 cylinder: diameter 32 cm, height 25 cm

17 rectangular prism: edges 3.1 m, 2.3 m, 1.8 m

19 cylinder: radius 1.4 m, height 2.3 m



C Riddle: What did the blanket say to the bed?

Calculate the volume of each prism (to the nearest cubic centimetre, where necessary). Then write the letter beside each prism above the correct answer.

C rectangular prism: edges 15 cm, 13 cm, 11 cm

D square prism: edges 9 cm, 9 cm, 14 cm

E cube: edges 10 cm

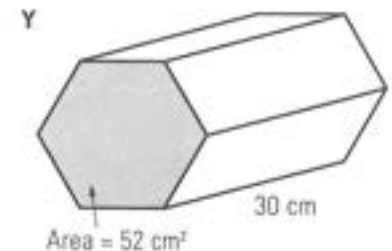
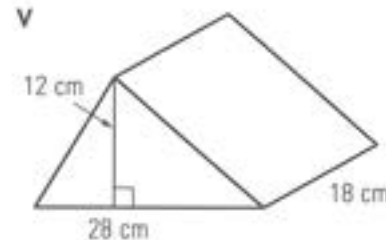
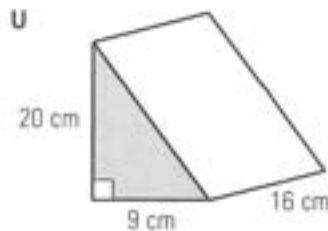
G rectangular prism: edges 18 cm, 16 cm, 9 cm

I rectangular prism: edges 20 cm, 17 cm, 12 cm

O cube: edges 15 cm

R cylinder: radius 9 cm, height 16 cm

T cylinder: diameter 15 cm, height 18 cm



4080	3024	1000	2592	3375	3181	1560	3375	1440	2145	3375	3024	1000	4072	1000	1134
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Note: all answers are cm^3

A

The **volume** of a pyramid is one-third the volume of a prism with the same base and perpendicular height.

$$V = \frac{1}{3} \text{ Area of base} \times \text{height}$$

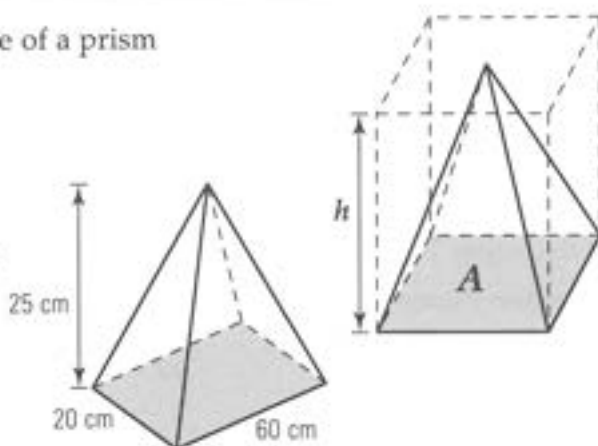
OR $V = \frac{1}{3} A h$

EXAMPLE:

A pyramid has a rectangular base 60 cm × 20 cm and perpendicular height 25 cm

$$\text{Area of base} = 60 \times 20 = 1\,200 \text{ cm}^2$$

$$\begin{aligned} \text{Volume} &= \frac{1}{3} \times 1\,200 \times 25 \\ &= 10\,000 \text{ cm}^3 \end{aligned}$$



Calculate the volume of each of these **rectangular pyramids**.

1 base: 30 cm × 15 cm
perpendicular height: 16 cm

2 base: 18 cm × 10 cm
perpendicular height: 14 cm

3 base: 20 cm × 16 cm
perpendicular height: 15 cm

4 base: 25 cm × 12 cm
perpendicular height: 18 cm

Calculate the volume of each of these **square pyramids** (i.e. pyramids with a square base).

5 base edge = 9 cm
perpendicular height = 14 cm

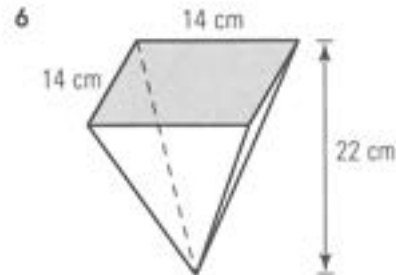
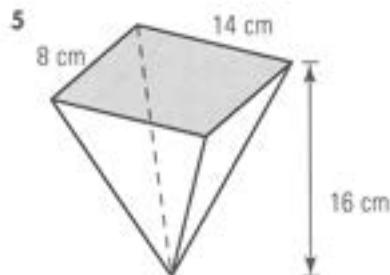
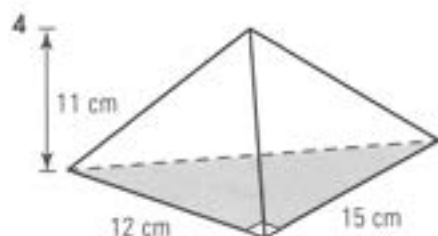
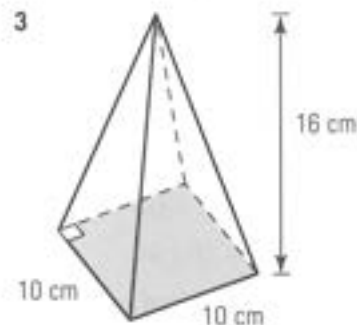
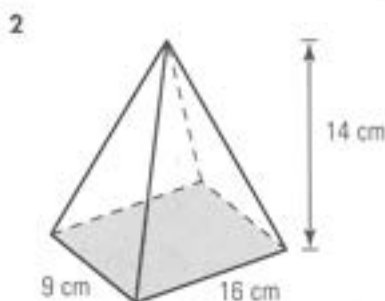
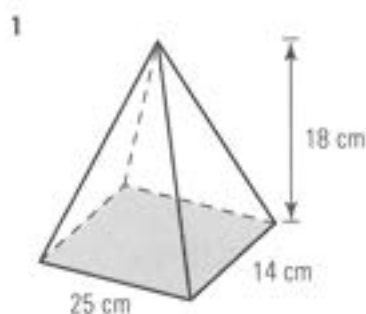
6 base edge = 12 cm
perpendicular height = 11 cm

7 base edge = 15 cm
perpendicular height = 10 cm

8 base edge = 11 cm
perpendicular height = 15 cm

B

Calculate the volume of each pyramid (correct to one decimal place, where necessary).



7 rectangular base: 6 m × 8.5 m
perpendicular height: 9.5 m

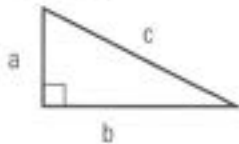
8 square base: edges 3.2 m
perpendicular height: 2.8 m

9 rectangular base: 4.8 m × 3.4 m
perpendicular height: 5.2 m

10 square base: edges 2.6 m
perpendicular height: 2.3 m

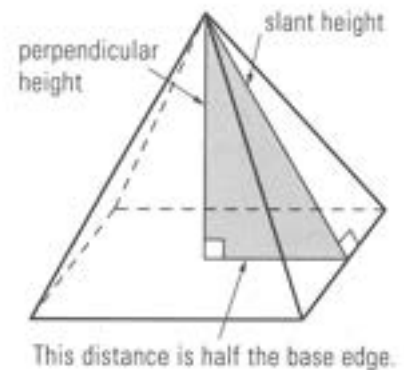
When you are given the **slant height** down one of the sloping faces of the pyramid, use **Pythagoras' theorem** to calculate the **perpendicular height**. You can then calculate the volume of the pyramid.

Pythagoras' theorem



In any right-angled triangle, the square on the hypotenuse is equal to the sum of the squares on the other two sides.

That is: $a^2 + b^2 = c^2$



Calculate the volume of each of these **square pyramids**.

11 base edge = 16 cm, slant height = 17 cm

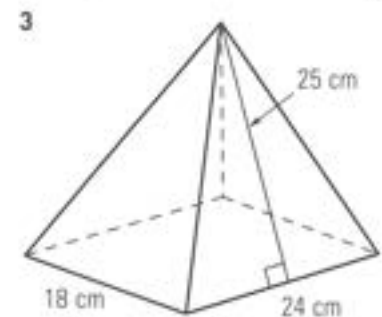
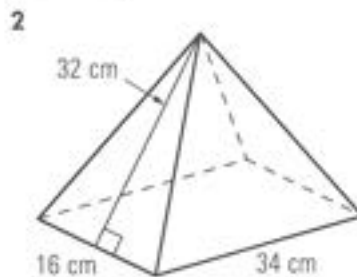
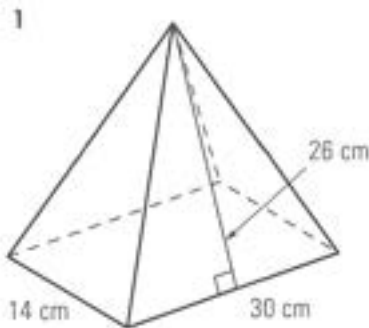
14 base edge = 18 cm, slant height = 15 cm

12 base edge = 24 cm, slant height = 20 cm

15 base edge = 32 cm, slant height = 34 cm

13 base edge = 20 cm, slant height = 26 cm

C Calculate the volume of each rectangular pyramid (correct to one decimal place, where necessary).



4 Calculate the volume of a triangular pyramid with base area of 40 cm^2 and perpendicular height of 27 cm.

6 A rectangular pyramid has a base $95 \text{ cm} \times 70 \text{ cm}$ and a volume of $99\,750 \text{ cm}^3$. What is the perpendicular height of the pyramid?

5 The base of a pyramid is an equilateral triangle with sides of 20 cm. The perpendicular height of the pyramid is 15 cm. Calculate the volume.

7 A square pyramid has a volume of 189 m^3 and a perpendicular height of 7 m. What is the length of each edge of its base?

Calculate the volume of each pyramid (correct to one decimal place, where necessary).

Hint: calculate the **perpendicular height** before finding the volume.

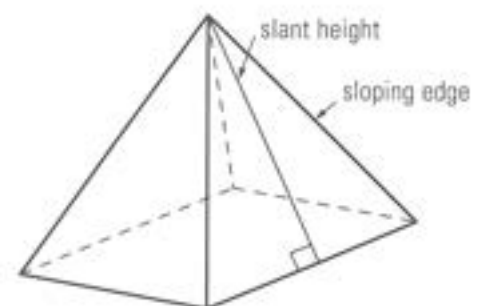
8 square base with edges 10 cm, sloping edges 12 cm

9 square base with edges 14 cm, sloping edges 18 cm

10 square base with edges 4.2 m, sloping edges 3.8 m

11 rectangular base $20 \text{ cm} \times 12 \text{ cm}$, sloping edges 18 cm

12 rectangular base $6.5 \text{ m} \times 4.3 \text{ m}$, sloping edges 7 m



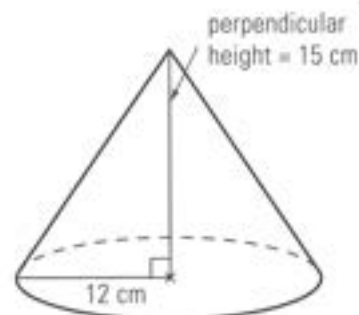
A

A cone is a pyramid with a circular base. The volume of a cone is calculated using the formula:

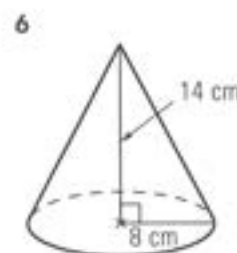
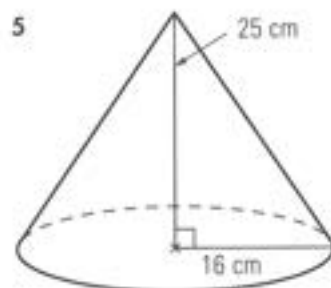
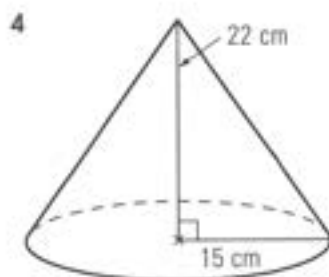
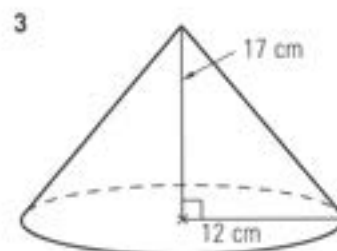
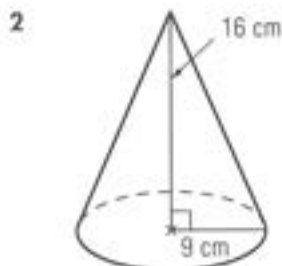
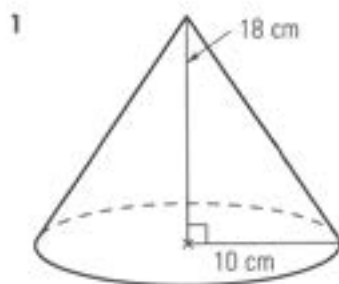
$$V = \frac{1}{3} A \times h \quad \text{where } A = \pi \times (\text{radius})^2$$

or $V = \frac{1}{3} \pi r^2 h$ Remember that h is the perpendicular height.

EXAMPLE: $\text{Volume} = \frac{1}{3} \times \pi \times 12^2 \times 15$
 $= 2261.9 \text{ cm}^3$
 (correct to one decimal place)

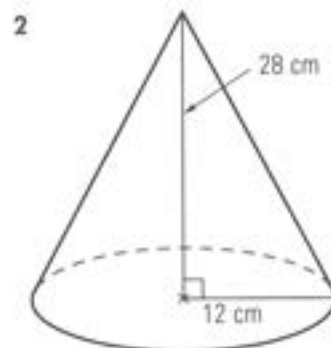
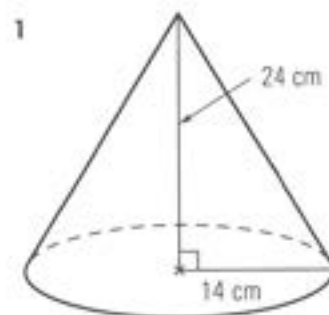


Calculate the volume of each cone (correct to one decimal place).



B

Calculate the volume of each cone (correct to one decimal place).



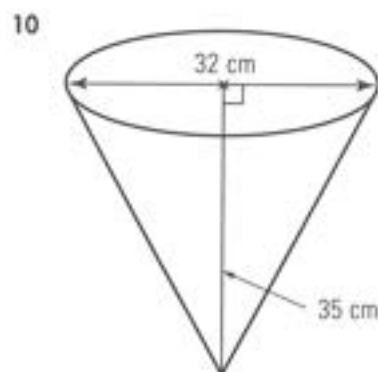
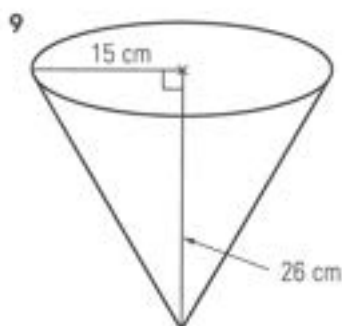
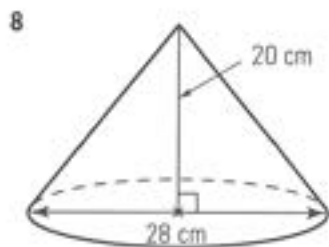
4 base radius = 12 cm
perpendicular height = 16 cm

6 base diameter = 28 cm
perpendicular height = 25 cm

5 base radius = 15 cm
perpendicular height = 18 cm

7 base diameter = 30 cm
perpendicular height = 20 cm

Calculate the volume of each cone (correct to one decimal place).



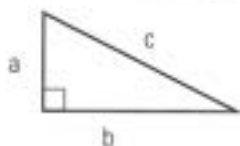
11 base radius = 11 cm
perpendicular height = 19 cm

12 base radius = 17 cm
perpendicular height = 24 cm

13 base diameter = 26 cm
perpendicular height = 24 cm

14 base diameter = 36 cm
perpendicular height = 27 cm

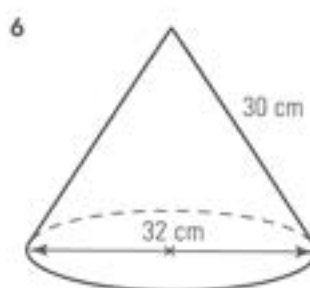
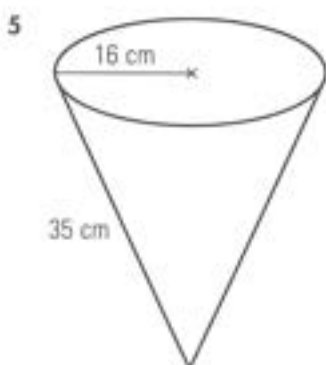
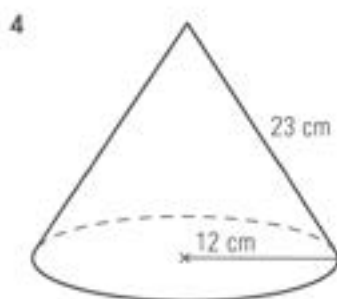
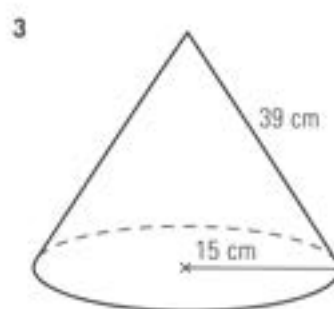
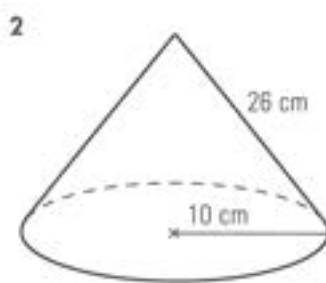
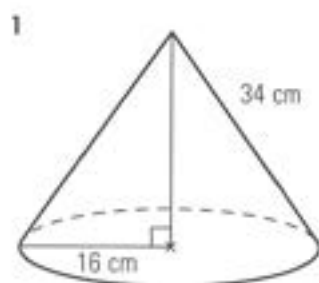
When you are given the slant height of the cone (down the curved surface), use **Pythagoras' theorem** to calculate the perpendicular height before you calculate the volume.



Pythagoras' theorem

In any right-angled triangle, the square on the hypotenuse is equal to the sum of the squares on the other two sides. That is: $a^2 + b^2 = c^2$

C Calculate the volume of each cone to the nearest cubic centimetre.



7 base radius = 17 cm, slant height = 33 cm

9 base diameter = 36 cm, slant height = 39 cm

8 base radius = 22 cm, slant height = 29 cm

10 base diameter = 35 cm, slant height = 38 cm

A

A **sphere** is a three-dimensional shape with one curved surface. All points on the surface of a sphere are the same distance from its centre. This distance is the **radius** of the sphere.

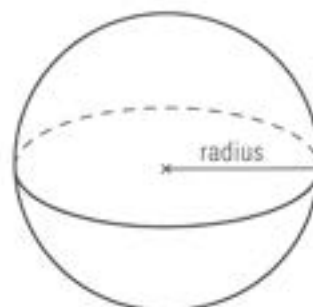
The volume of a sphere with radius r units is given by:

$$V = \frac{4}{3} \pi r^3$$

OR $V = \frac{4}{3} \times \pi \times (\text{radius})^3$

EXAMPLE: A sphere with radius 8 cm

$$\begin{aligned} \text{Volume} &= \frac{4}{3} \times \pi \times 8^3 \\ &= 2144.7 \text{ cm}^3 \\ &\text{(correct to one decimal place)} \end{aligned}$$



Use the given radius to calculate the volume of each sphere (correct to one decimal place).

1 radius = 10 cm

2 radius = 9 cm

3 radius = 15 cm

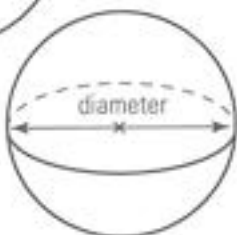
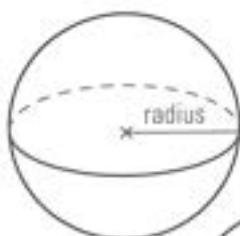
4 radius = 12 cm

5 radius = 6 cm

6 radius = 14 cm

B

Use the given dimension to calculate the volume of each sphere (correct to one decimal place).



1 radius = 18 cm

2 radius = 25 cm

3 radius = 1.6 m

4 radius = 3.5 m

5 radius = 0.9 m

6 radius = 2.3 m

7 radius = 65 mm

8 radius = 57 mm

9 diameter = 16 cm

10 diameter = 22 cm

11 diameter = 26 cm

12 diameter = 25 cm

13 diameter = 35 mm

14 diameter = 1.2 m

15 diameter = 1.6 m

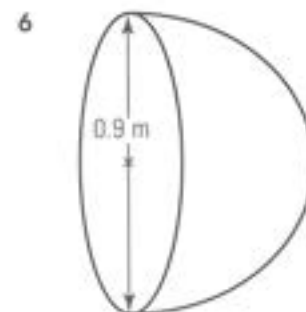
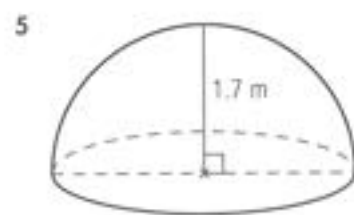
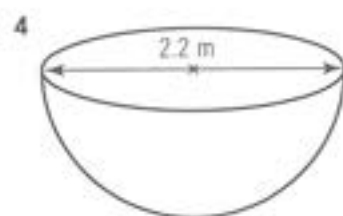
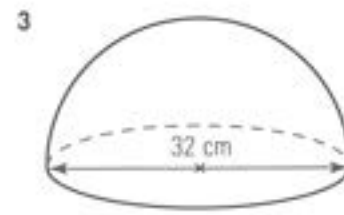
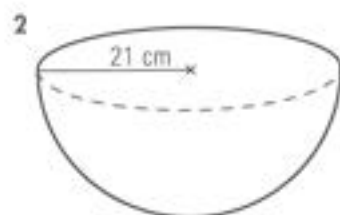
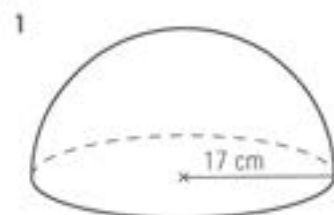
16 diameter = 2.5 m

17 diameter = 60 mm

18 diameter = 74 mm

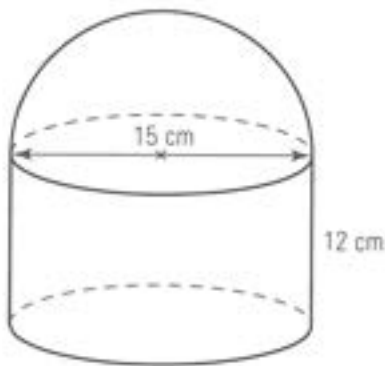
C

Calculate the volume of each hemisphere or half-sphere (correct to two decimal places).

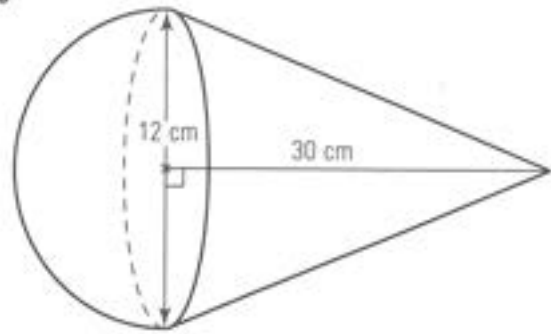


Calculate the total volume of each solid (correct to one decimal place).

7



8



- 9 The volume of a sphere is 8180 cm^3 . Calculate its diameter correct to one decimal place.

- 10 A container carrying liquified gas is in the shape of a cylinder 5 m long together with two hemispherical ends. The total length of the container is 6.8 m. What is its volume in cubic metres (correct to one decimal place)?

The **volume** of a sphere can be used to calculate its **radius**.

EXAMPLE: Calculate the radius of a sphere with volume 2150 cm^3 .

$$\text{Volume} = \frac{4}{3} \pi r^3$$

$$2150 = \frac{4}{3} \times \pi \times r^3$$

$$\begin{aligned} \text{rearrange: } r^3 &= \frac{2150 \times 3}{4 \times \pi} \\ &= 513.274 \dots \end{aligned}$$

Obtain the cube root: $r = 8.0 \text{ cm}$ (to one decimal place)

Calculate the **radius** of each sphere, correct to one decimal place.

11 volume = 2000 cm^3

12 volume = 1600 cm^3

13 volume = 1850 cm^3

14 volume = 2400 cm^3

15 volume = 2550 cm^3

16 volume = 3260 cm^3

17 volume = 9.5 m^3

18 volume = 11.4 m^3

19 volume = 15.3 m^3

Calculate the **diameter** of each sphere, correct to one decimal place.

20 volume = 2280 cm^3

21 volume = 1960 cm^3

22 volume = 2740 cm^3

23 volume = 10.6 m^3

24 volume = 8.7 m^3

25 volume = 14.8 m^3

Volume formulae

prism:

$$V = A h$$

pyramid: $V = \frac{1}{3} A h$

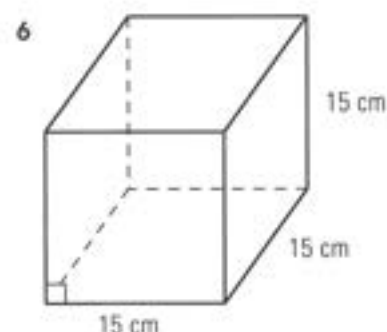
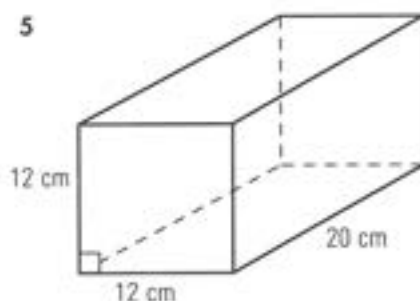
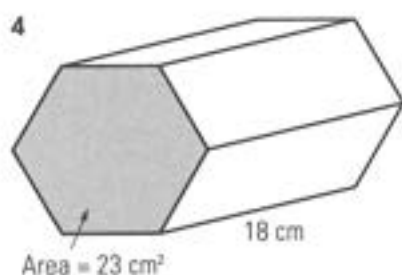
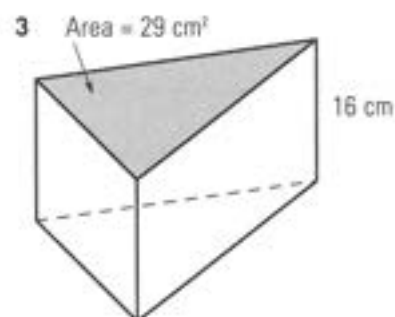
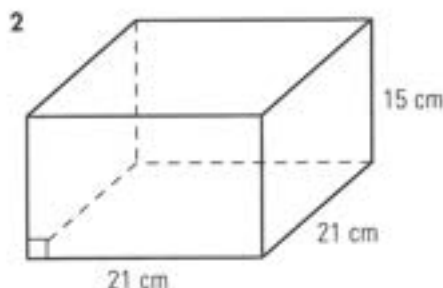
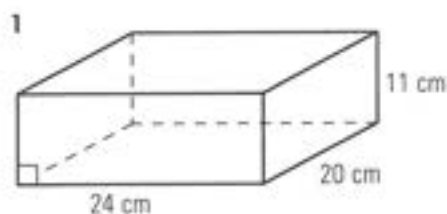
rectangular prism: $V = l \times b \times h$

cone: $V = \frac{1}{3} \pi r^2 h$

cylinder: $V = \pi r^2 h$

sphere: $V = \frac{4}{3} \pi r^3$

A Calculate the volume of each prism.



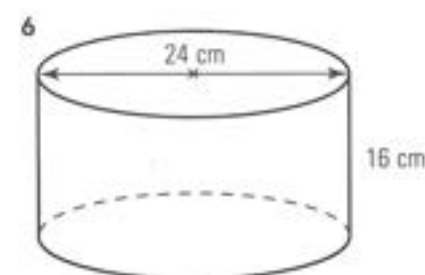
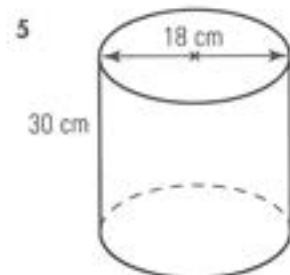
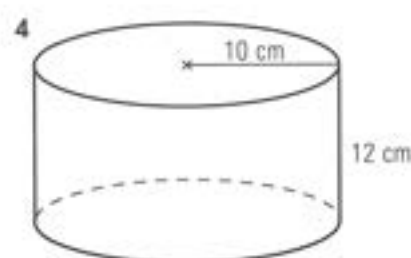
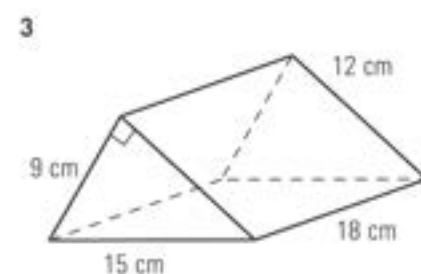
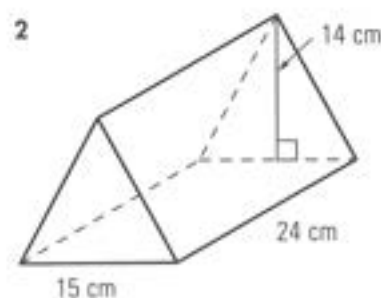
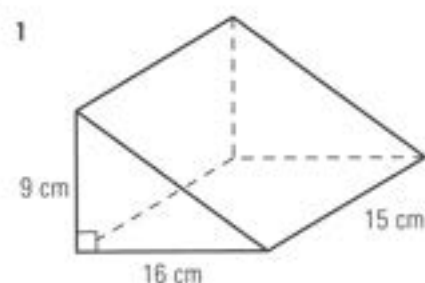
7 rectangular prism: edges 16 cm, 14 cm, 11 cm

8 rectangular prism: edges 22 cm, 15 cm, 26 cm

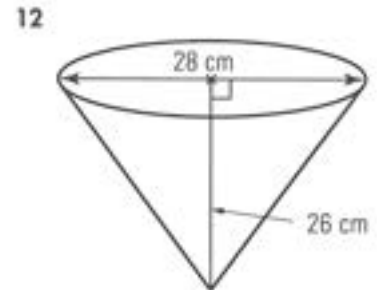
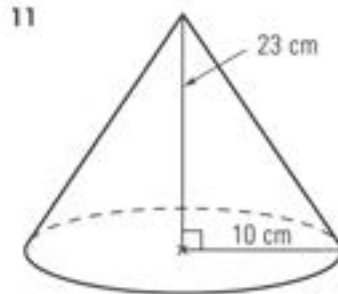
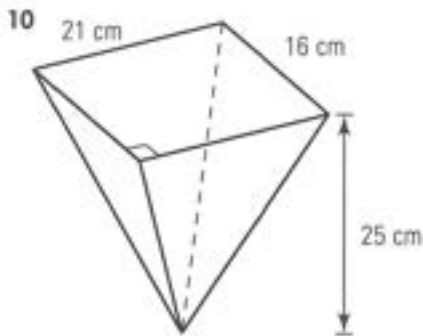
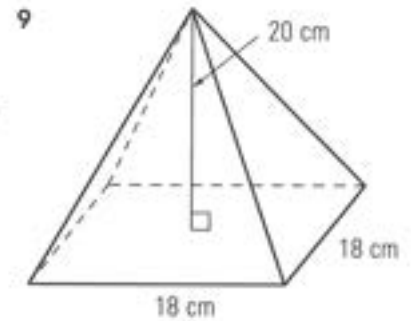
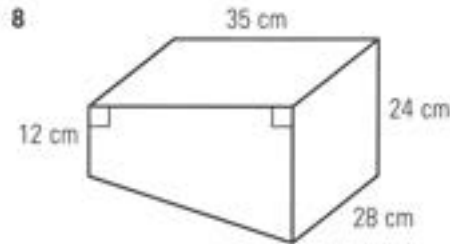
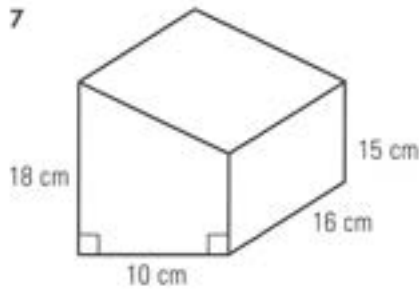
9 cube: edges 18 cm

10 square prism: edges 14 cm, 14 cm, 17 cm

B Calculate the volume of each solid shape (to one decimal place, where necessary).



Calculate the volume of each solid shape (to one decimal place, where necessary).



13 cylinder: radius 9 cm, height 17 cm

15 rectangular pyramid: base 14 cm $\times$ 8 cm, perpendicular height 18 cm

17 cone: base diameter 24 cm, height 32 cm

19 sphere: radius 11.5 cm

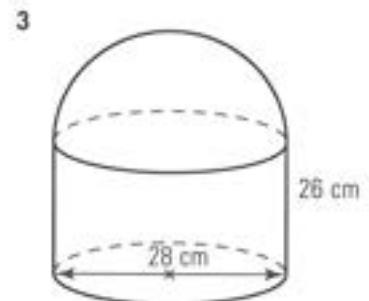
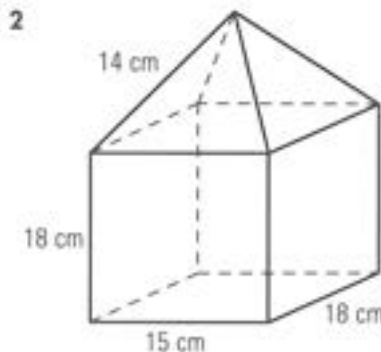
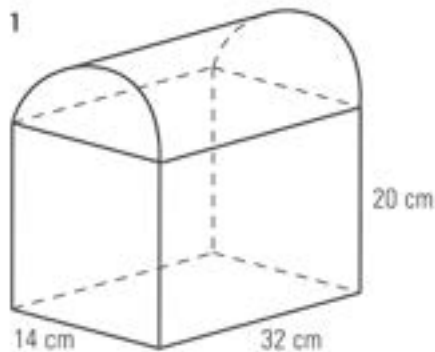
14 cone: base radius 25 cm, height 36 cm

16 square pyramid: base edge 24 cm, perpendicular height 27 cm

18 sphere: radius 8 cm

20 sphere: diameter 27 cm

C Calculate the total volume of each solid (to one decimal place, where necessary).



4 A water tank is the shape of a rectangular prism, length 1.9 m and breadth 0.8 m. It contains water to a depth of 65 cm. How many litres of water are in the tank? (Hint: 1 litre occupies 1000 cm³.)

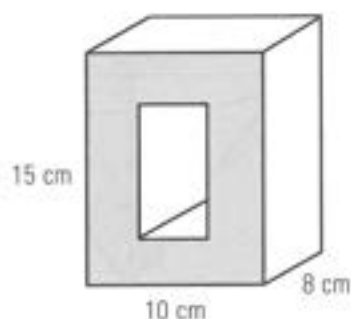
6 What space would remain in a cubical box, each edge 30 cm, if a ball of diameter 25 cm was placed inside it?

5 Two cylindrical containers have been labelled A and B. Container A is 6 cm high and has a diameter of 20 cm. Container B is 20 cm high and has a diameter of 6 cm. Which container will hold the most? What is the difference of their volumes?

7 A rectangular pyramid has a base measuring 9 m by 7 m and a volume of 252 m³. What is its perpendicular height?

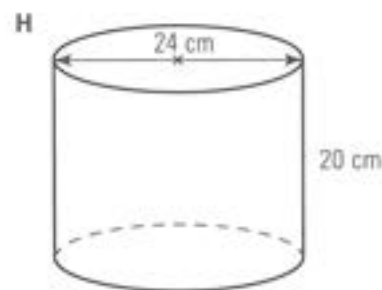
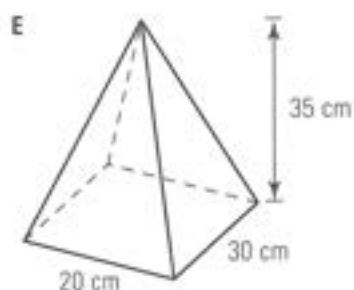
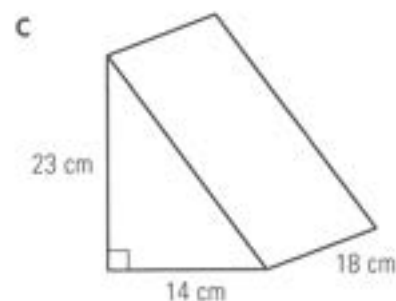
Calculate the total volume of each solid (to one decimal place, where necessary).

- 8 A square pyramid has a volume of 189 m^3 and a perpendicular height of 7 m. What is the length of the edges of its base?
- 9 A cone with a diameter of 42 cm has a volume of $23\,100 \text{ cm}^3$. Calculate its perpendicular height, to the nearest centimetre.
- 10 A cone has a diameter of 32 cm and a slant height of 34 cm. Calculate its volume.
- 11 The cross-section of a metal pipe is an annulus (two concentric circles). The inner diameter is 14 cm, the outer diameter is 18 cm. Calculate the volume of the metal in a 2.5-metre length of this pipe.



Riddle: What is a pig's favourite karate chop?

Calculate the volume of each solid (to the nearest cubic centimetre, where necessary). Then write the letter beside each solid above each correct answer.



K rectangular prism: edges 14 cm, 16 cm, 9 cm

P cylinder: diameter 24 cm, height 16 cm

T sphere: diameter 16 cm

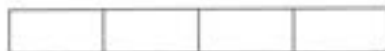
O cube: edges 14.5 cm

R cone: base radius 12 cm, perpendicular height 15 cm

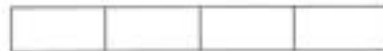
Note: all answers are cm^3



2144.7 9047.8 7000



7238.2 3048.6 2261.9 2016



2898 9047.8 3048.6 7238.2

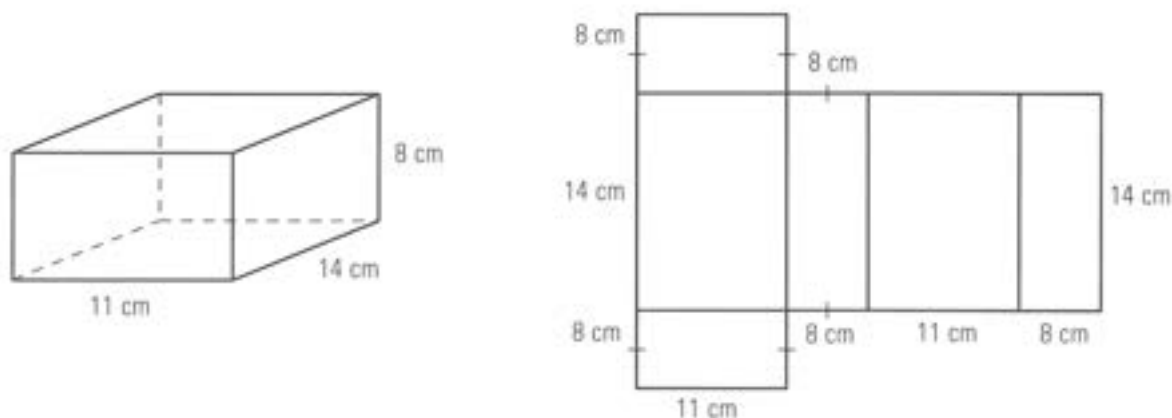
A

The surface area of a solid = the sum of the areas of its faces.

A rectangular prism has six faces. Drawing the net of a prism can help with calculating the surface area.

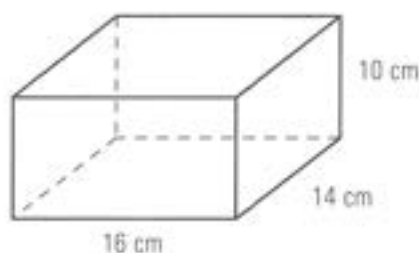
A net is a flat diagram that consists of the faces of a solid shape. The faces are arranged so that the diagram could be folded up to form that solid.

- The net of this rectangular prism has been drawn and labelled for you. Calculate the surface area of the prism by finding the sum of the areas of its faces.

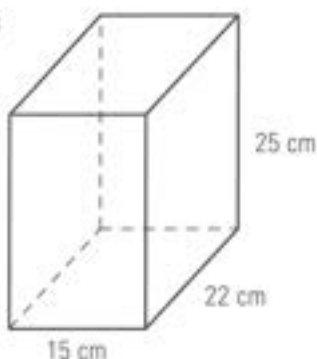


For each rectangular prism, draw and label its net and then calculate the surface area.

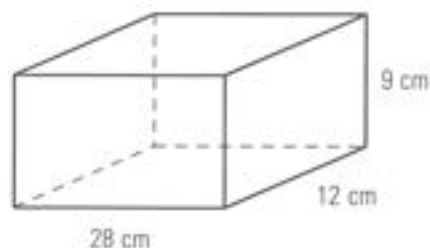
2



3

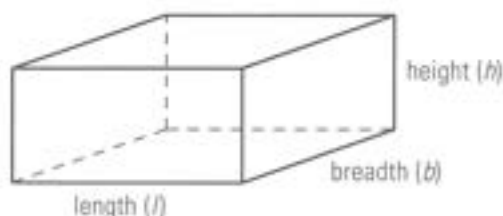


4

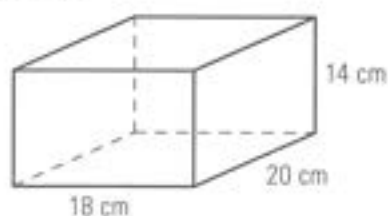


You may wish to learn the following formula for the surface area (SA) of a rectangular prism.

$$SA = 2(lb + lh + bh)$$

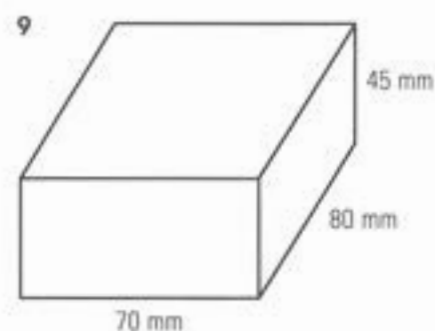
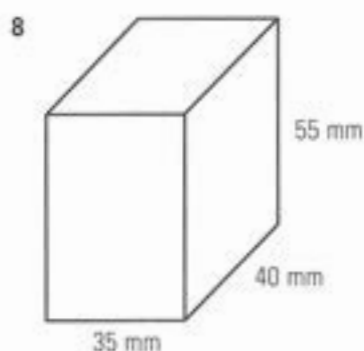
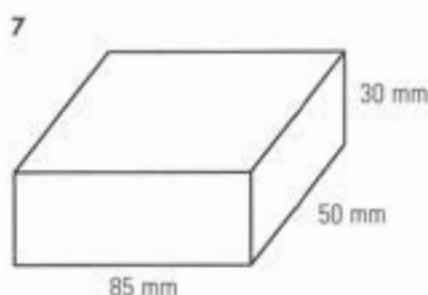
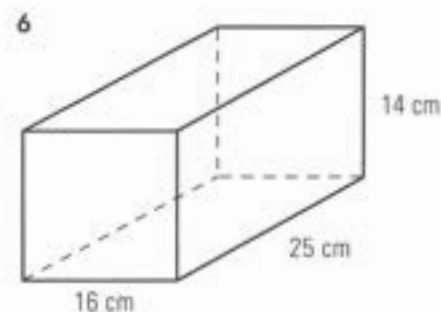
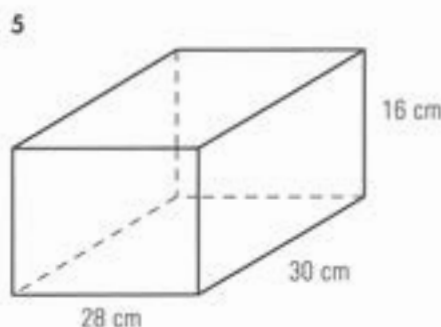
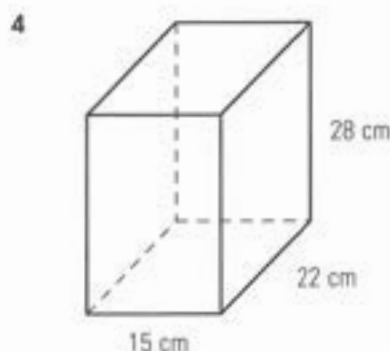
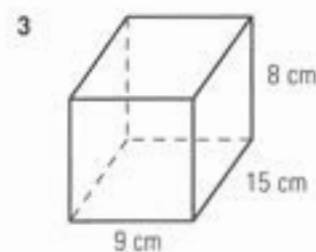
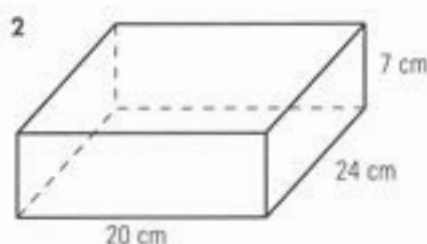
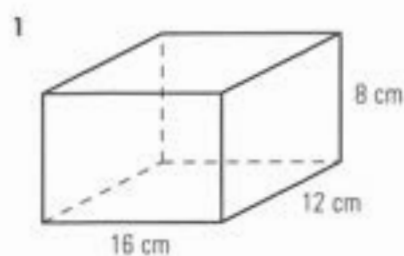


EXAMPLE:



$$SA = 2 \times (18 \times 20 + 18 \times 14 + 20 \times 14) = 1784 \text{ cm}^2$$

B Calculate the surface area of each rectangular prism.



C Use the given dimensions to calculate the surface area of each rectangular prism.

1 length = 18 cm
breadth = 14 cm
height = 12 cm

2 length = 16 cm
breadth = 13 cm
height = 11 cm

3 length = 22 cm
breadth = 16 cm
height = 15 cm

4 length = 20 cm
breadth = 14 cm
height = 22 cm

5 length = 80 mm
breadth = 45 mm
height = 25 mm

6 length = 90 mm
breadth = 40 mm
height = 55 mm

7 length = 85 mm
breadth = 50 mm
height = 65 mm

8 length = 3.5 m
breadth = 2.5 m
height = 1.8 m

9 length = 5.6 m
breadth = 3.2 m
height = 2.5 m

10 length = 4.8 m
breadth = 1.6 m
height = 3.3 m

11 length = 3.2 m
breadth = 2.7 m
height = 1.5 m

12 length = 4.3 m
breadth = 2.4 m
height = 3.5 m

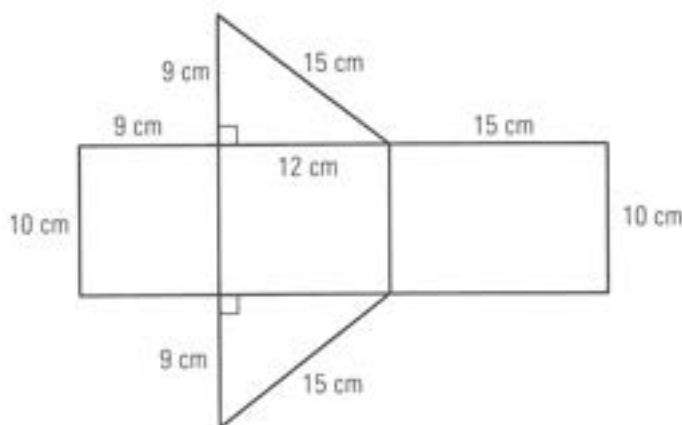
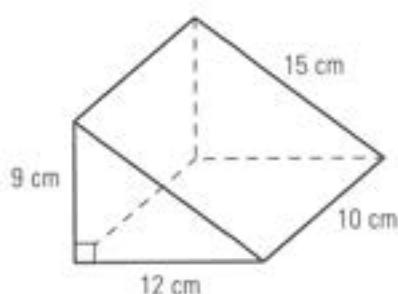
A

A **triangular prism** has five faces. They are two congruent triangles (the two ends) and three rectangles. Its cross-section is a triangle.

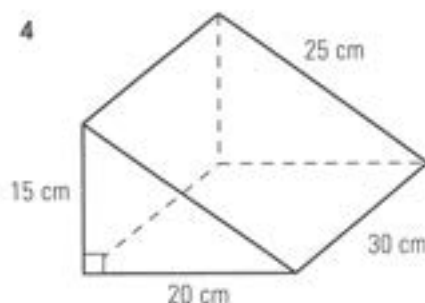
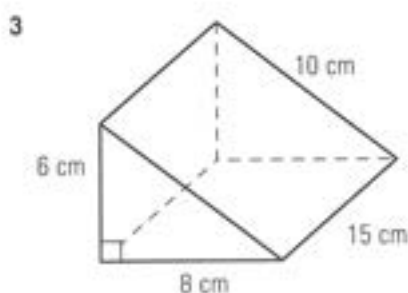
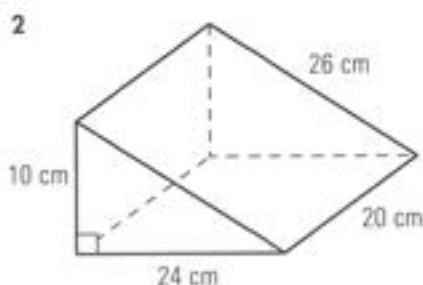
The **surface area** of a solid = the sum of the areas of its faces.

A **net** is a flat diagram that consists of the faces of a solid shape. The faces are arranged so that the diagram could be folded up to form that solid. Drawing the net of a prism can help with calculating the surface area.

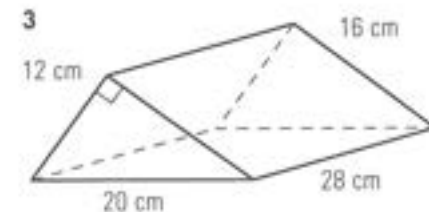
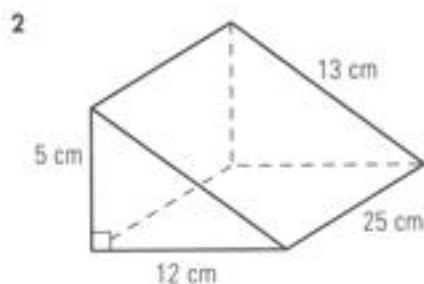
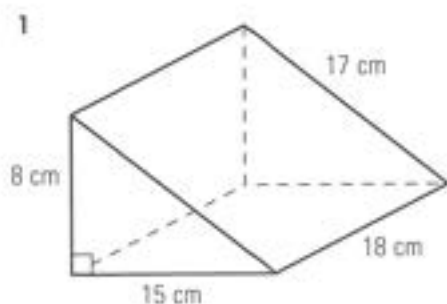
- 1 The diagram shows the net of this triangular prism. Calculate its surface area.



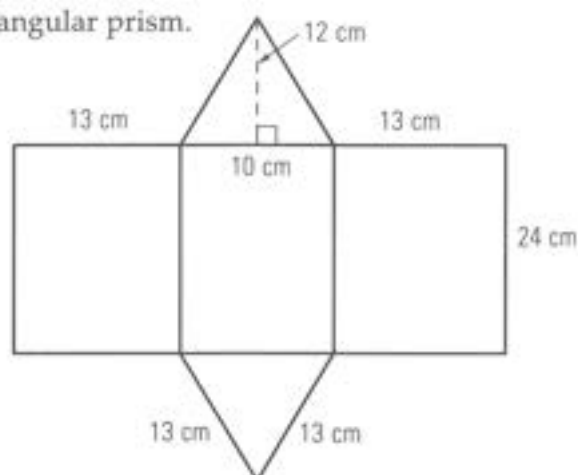
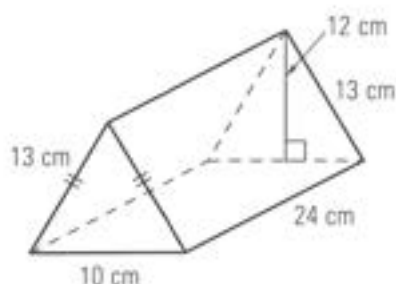
For each triangular prism, sketch a diagram of the net and then calculate the surface area.



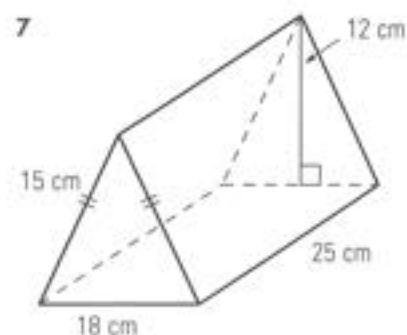
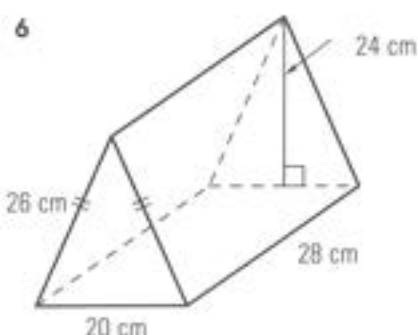
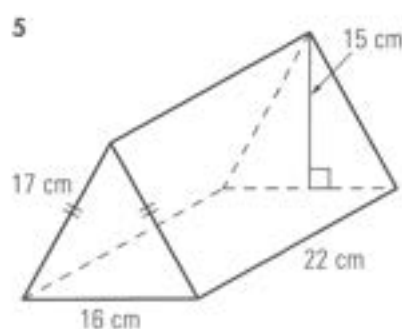
- B** Calculate the surface area of each rectangular prism.



- 4 The diagram shows the net (not to scale) of this triangular prism. The triangular faces are isosceles triangles. Calculate the surface area of this solid.



Calculate the surface area of each of these triangular prisms.

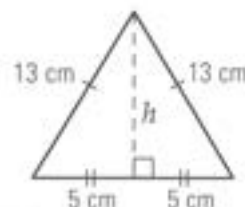


C

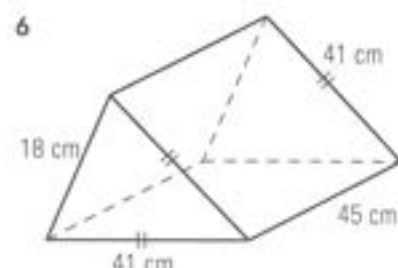
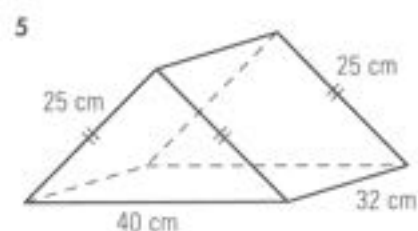
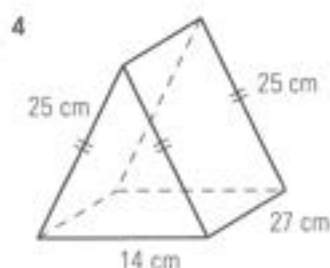
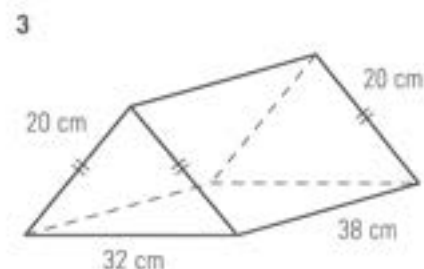
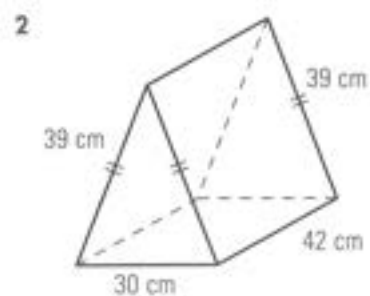
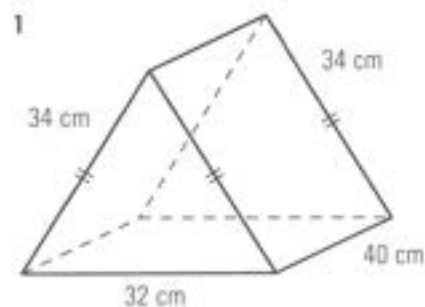
In an isosceles triangle, the axis of symmetry can be used to calculate the height of the triangle. The axis of symmetry forms two right-angled triangles. By applying **Pythagoras' theorem**, you can calculate the height.

EXAMPLE:

$$\begin{aligned} h^2 &= 13^2 - 5^2 \\ &= 144 \\ h &= \sqrt{144} \\ &= 12 \text{ cm} \end{aligned}$$



Calculate the surface area of each triangular prism. (Hint: first use Pythagoras' theorem to calculate the height.)

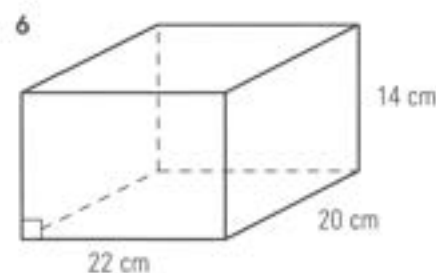
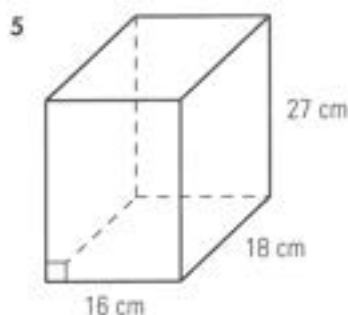
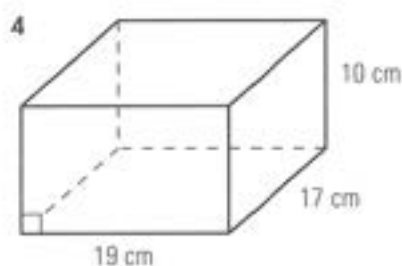
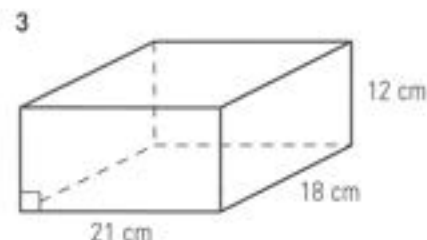
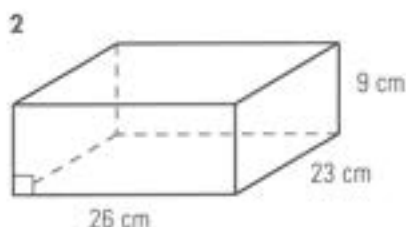
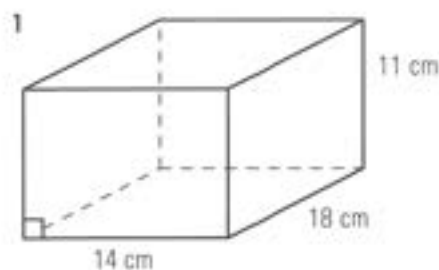


A

The **surface area** of a solid = the sum of the areas of its faces.

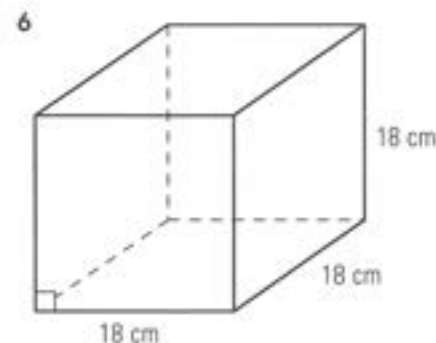
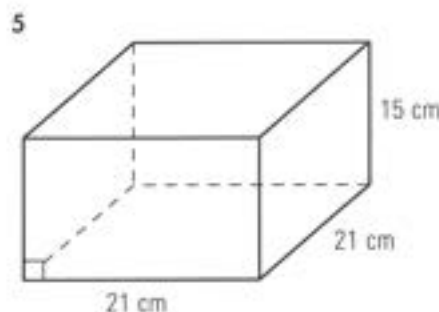
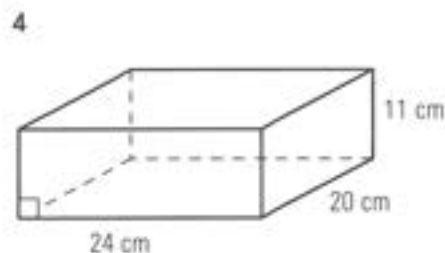
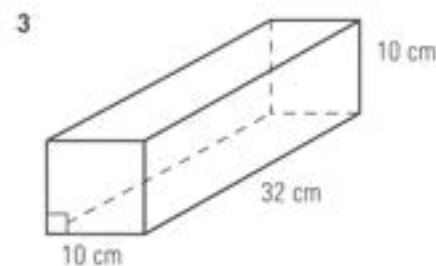
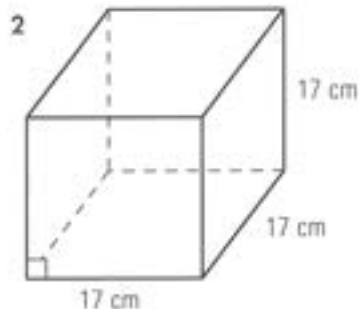
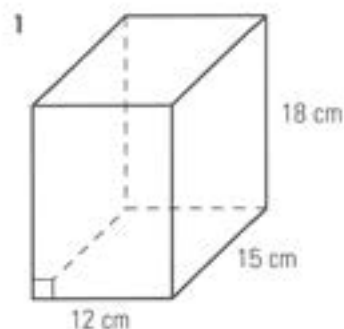
A prism is a solid with uniform cross-section. It has two opposite faces which are parallel and congruent (same size and same shape).

Calculate the surface area of each of these rectangular prisms.

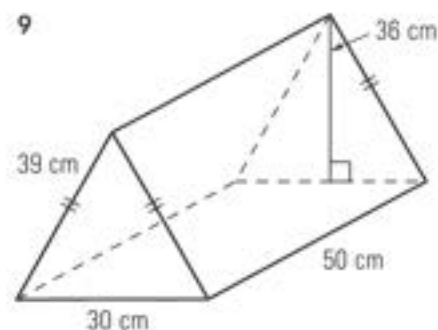
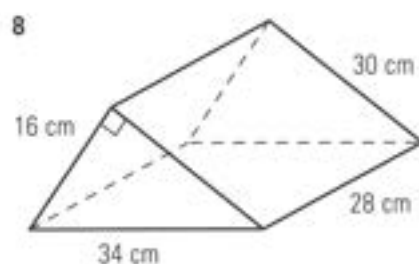
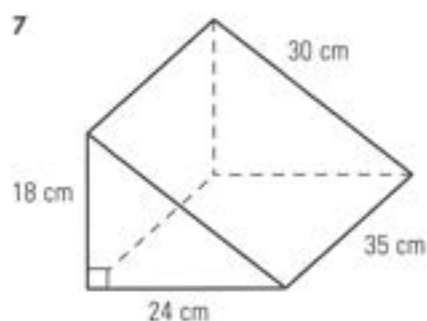


B

Calculate the surface area of each prism.
Remember to calculate the area of each face of the solid.



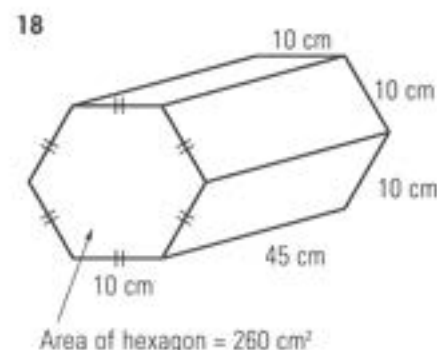
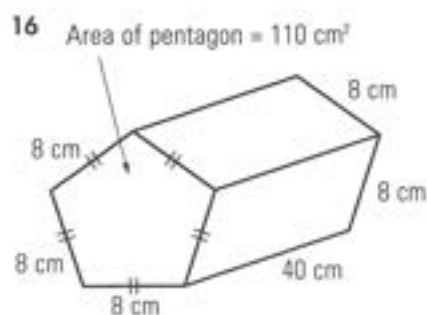
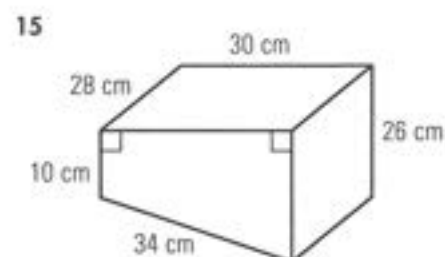
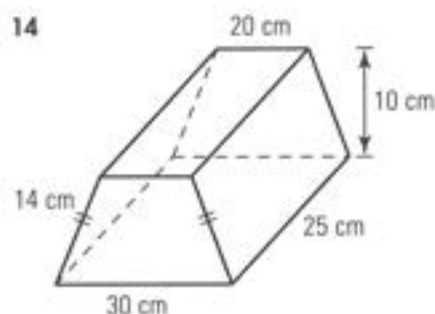
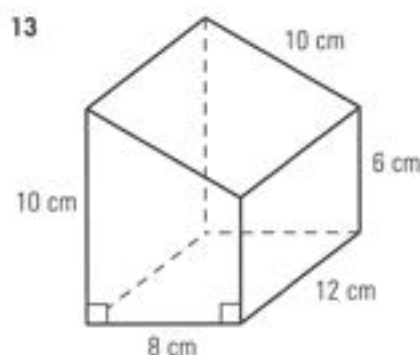
Calculate the surface area of each prism.
Remember to calculate the area of each face of the solid.



10 A rectangular prism with edges 28 cm, 24 cm, 19 cm

11 A cube with edges 15 cm

12 A square prism with edges 16 cm, 16 cm, 25 cm



19 rectangular prism: edges 2.3 m, 1.4 m, 1.3 m

20 rectangular prism: edges 3.8 m, 2.6 m, 1.9 m

21 cube: edges 2.5 m

22 square prism: edges 2.6 m, 2.6 m, 3.5 m

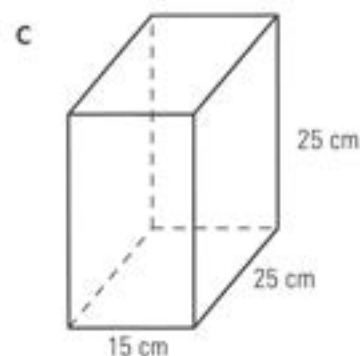
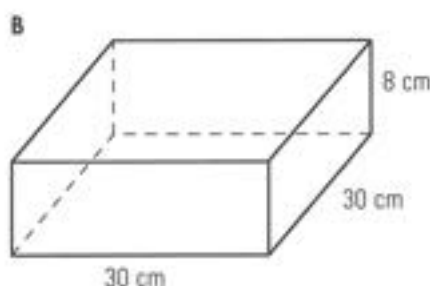
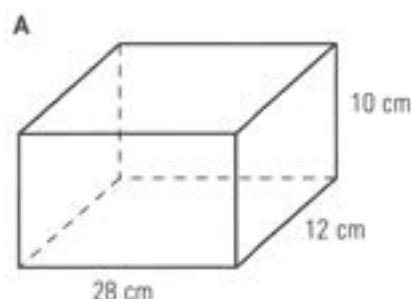
23 square prism: edges 1.5 m, 1.5 m, 2.9 m

24 rectangular prism: edges 4.2 m, 2.4 m, 2.1 m

C Riddle: Why did the jellybean go to school?

Calculate the surface area of each solid.

Then write the letter beside each solid above the correct answer.



D cube: edges 30 cm

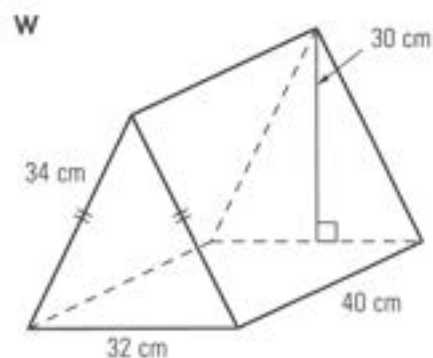
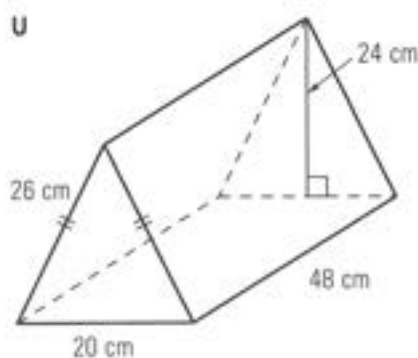
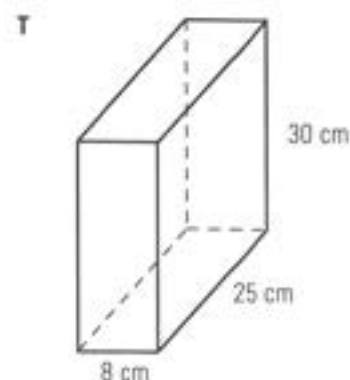
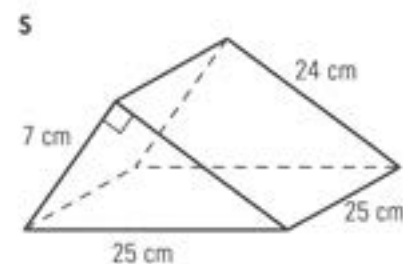
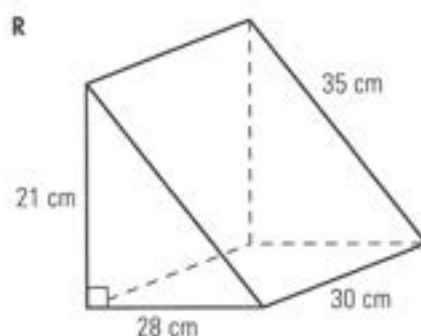
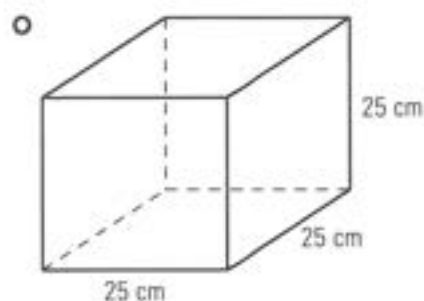
E rectangular prism: edges 35 cm, 22 cm, 18 cm

H rectangular prism: edges 24 cm, 17 cm, 29 cm

I square prism: edges 15 cm, 15 cm, 25 cm

M square prism: edges 20 cm, 20 cm, 30 cm

N cube: edges 15 cm



Note: all answers are cm^2

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2760 3592 2750 1472 3936 1568 3592

--	--

3194 3592

--	--	--	--	--	--	--	--

4960 1472 1350 2380 3592 5400

--	--

2380 3750

--	--	--	--	--	--	--	--

2760 3592 2750 3750 3200 3592

--

1472

--	--	--	--	--	--	--	--	--	--

1568 3200 1472 3108 2380 1950 3592

A

A cylinder has two congruent faces (circles at each end) and a curved surface.

A solid cylinder is closed at both ends and is often called a **closed cylinder**.

Total surface area of a cylinder = area of curved surface + area of 2 circular ends

$$SA = 2\pi rh + 2\pi r^2$$

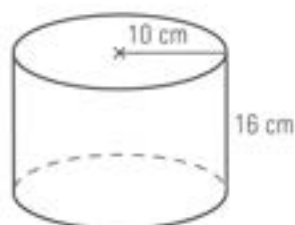
EXAMPLE:

The total surface area of this closed cylinder

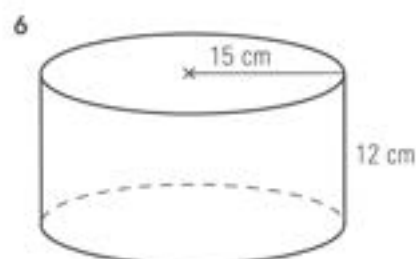
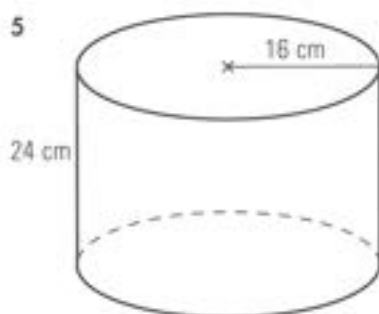
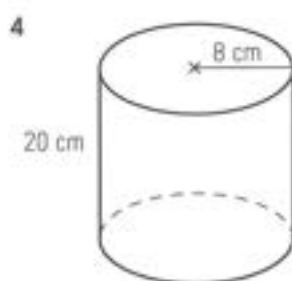
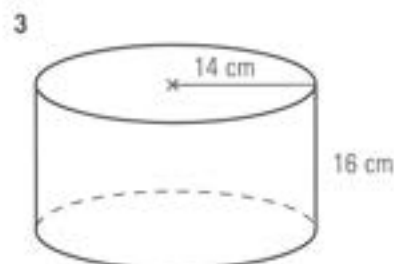
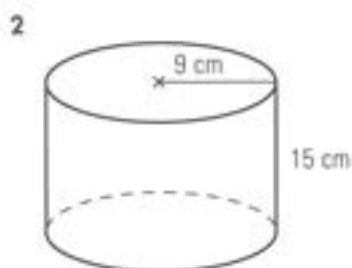
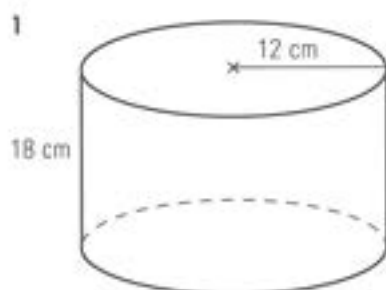
= area of curved surface + area of 2 ends

$$= 2 \times \pi \times 10 \times 16 + 2 \times \pi \times 10^2$$

$$= 1633.6 \text{ cm}^2 \text{ (correct to one decimal place)}$$

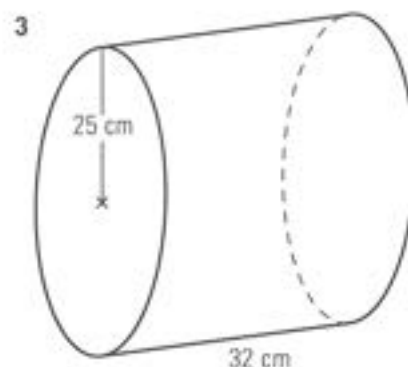
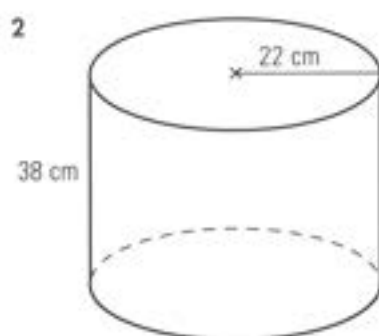
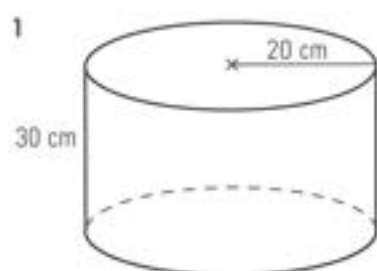


Calculate the total surface area of each of these closed cylinders, correct to one decimal place.

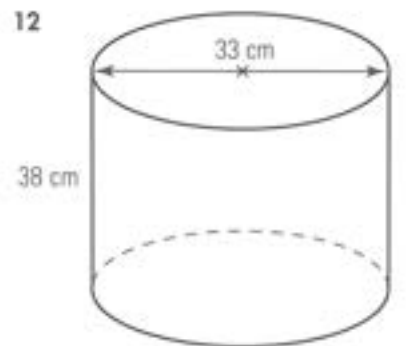
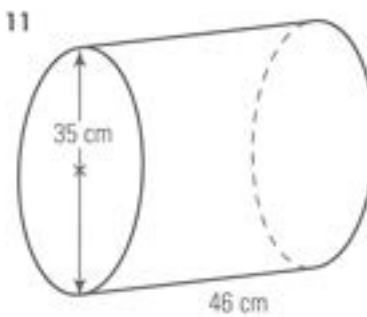
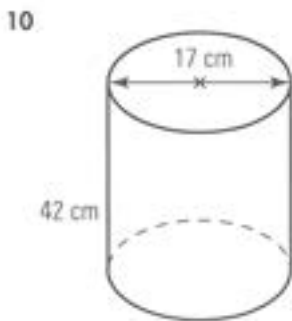
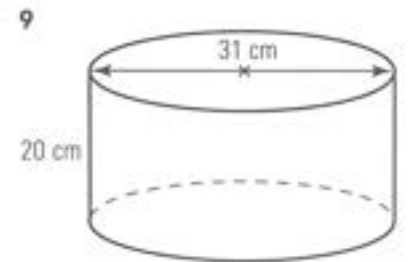
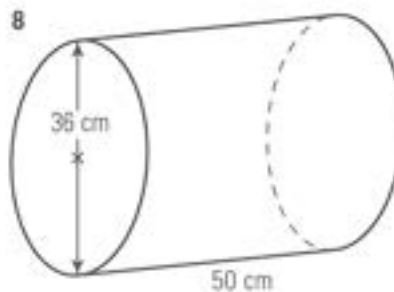
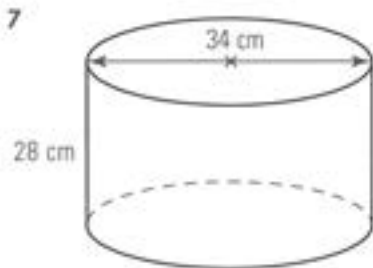
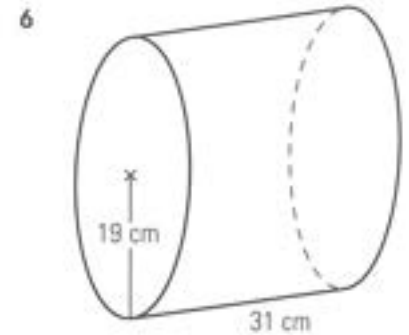
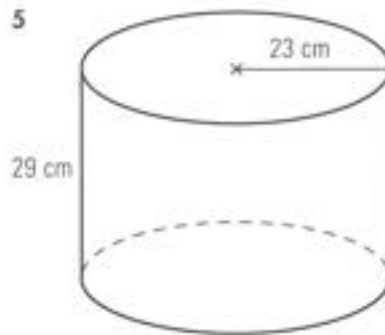
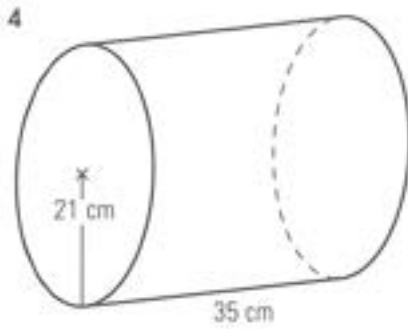


B

Calculate the total surface area of each closed cylinder, correct to one decimal place.



Calculate the total surface area of each closed cylinder, correct to one decimal place.



13 cylinder: radius 1.8 m, height 2.6 m

15 cylinder: radius 1.3 m, height 3.05 m

17 cylinder: radius 105 mm, height 360 mm

19 cylinder: height 5.6 m, diameter 3.4 m

21 cylinder: height 2.9 m, diameter 1.6 m

23 cylinder: 85 mm in diameter, 260 mm high

14 cylinder: radius 0.9 m, height 1.4 m

16 cylinder: radius 85 mm, height 250 mm

18 cylinder: radius 56 mm, height 140 mm

20 cylinder: height 4.8 m, diameter 6.5 m

22 cylinder: 105 mm in diameter, 200 mm high

24 cylinder: 90 mm in diameter, 385 mm high

G

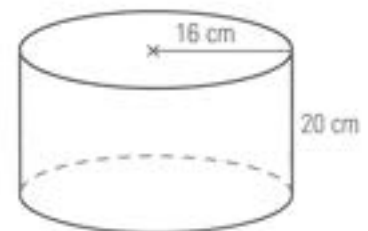
EXAMPLE:

The area of the curved surface around this cylinder

$$= 2\pi rh$$

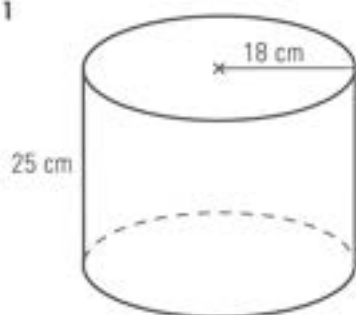
$$= 2 \times \pi \times 16 \times 20$$

$$= 2010.6 \text{ cm}^2 \text{ (correct to one decimal place)}$$

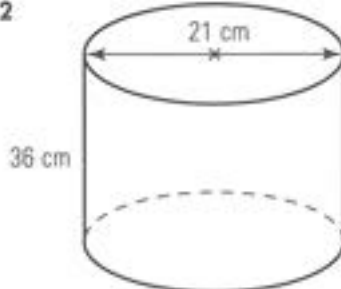


Calculate the area of the curved surface around these cylinders (correct to one decimal place).

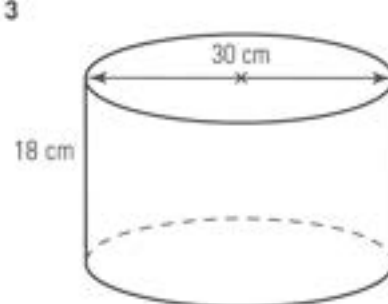
1



2



3

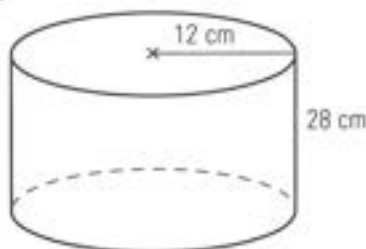


4 cylinder: 38 cm in diameter, 40 cm high

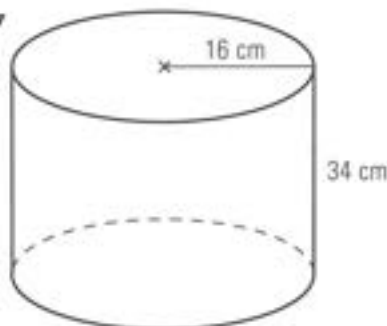
5 cylinder: 55 cm in diameter, 65 cm high

These cylinders are **open** at one end. Calculate the **external surface area** (base and curved surface) of each cylinder, correct to one decimal place.

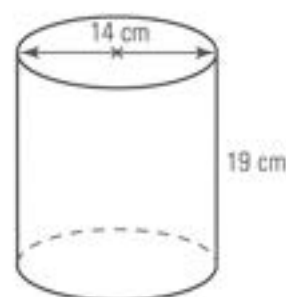
6



7



8

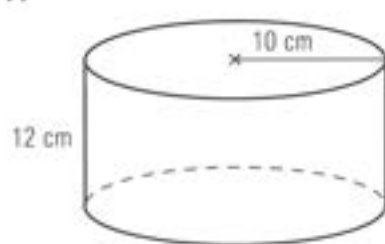


9 open cylinder: 11 cm in diameter, 18 cm high

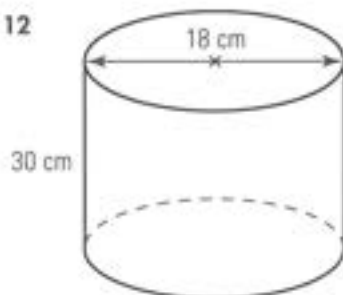
10 open cylinder: 32 cm in diameter, 25 cm high

These cylinders are **open** at one end. Calculate the **total surface area** (inside and outside) of each cylinder correct to one decimal place.

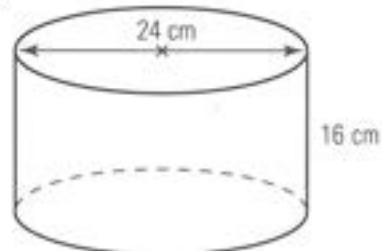
11



12



13

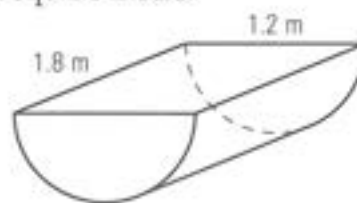
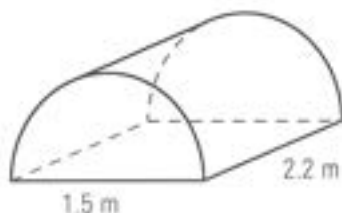


14 open cylinder: 140 mm in diameter, 125 mm high

15 open cylinder: 115 mm in diameter, 130 mm high

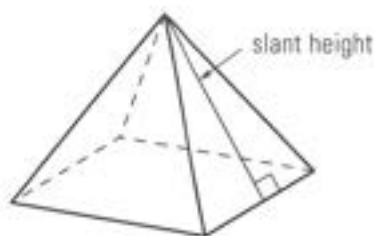
16 Calculate the total surface area of this closed half cylinder, to the nearest square metre.

17 This water trough is a half cylinder open at the top. Calculate the outside surface area to the nearest square metre.

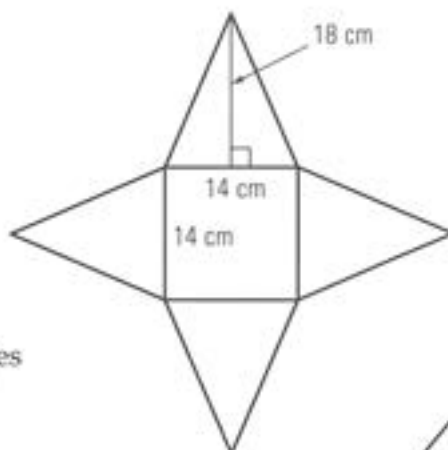


A

A **square pyramid** has a square base and four congruent triangular faces.



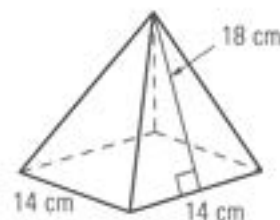
These diagrams show a square pyramid and the net of this square pyramid. The net consists of a square and four congruent triangles.



EXAMPLE:

The surface area of this pyramid

$$\begin{aligned}
 &= \text{area of base} + \text{area of 4 triangles} \\
 &= 14 \times 14 + 4 \times \frac{1}{2} \times 14 \times 18 \\
 &= 196 + 504 \\
 &= 700 \text{ cm}^2
 \end{aligned}$$



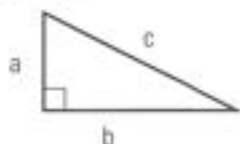
Calculate the total surface area of each of these square pyramids.

- 1 base edge = 15 cm, slant height = 16 cm
- 2 base edge = 20 cm, slant height = 18 cm
- 3 base edge = 18 cm, slant height = 21 cm
- 4 base edge = 25 cm, slant height = 24 cm

B

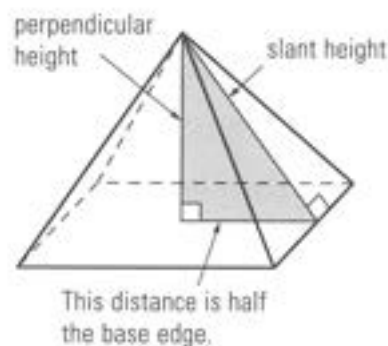
When you are given the **perpendicular height** of a pyramid, use **Pythagoras' theorem** to calculate the slant height. You can then calculate the surface area of the pyramid.

Pythagoras' theorem



In any right-angled triangle, the square on the hypotenuse is equal to the sum on the squares of the other two sides.

That is: $a^2 + b^2 = c^2$



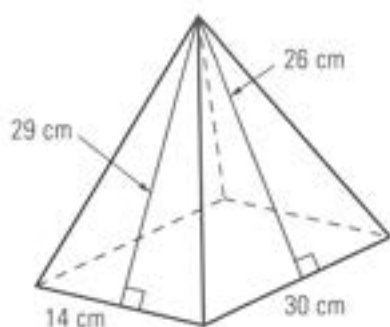
Calculate the total surface area of each of these square pyramids.

- 1 base edge = 24 cm
perpendicular height = 16 cm
- 2 base edge = 32 cm
perpendicular height = 30 cm
- 3 base edge = 20 cm
perpendicular height = 24 cm
- 4 base edge = 18 cm
perpendicular height = 12 cm

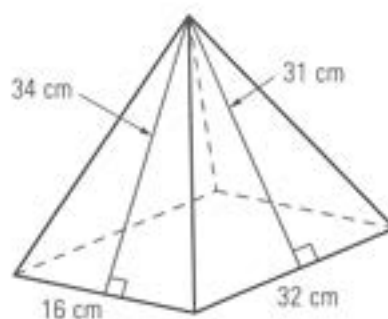
In a **rectangular pyramid**, the sloping faces are not all the same size. However, the opposite triangular faces are congruent.

Calculate the total surface area of each of these rectangular pyramids, correct to one decimal place.

5



6



7 base length = 20 cm, breadth = 16 cm, perpendicular height = 20 cm

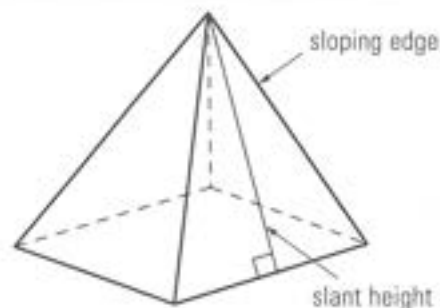
9 rectangular base 18 cm $\times$ 10 cm, perpendicular height = 16 cm

8 base length = 24 cm, breadth = 18 cm, perpendicular height = 25 cm

10 rectangular base 6 m $\times$ 8.5 m, perpendicular height = 9.5 m

C

When you are given the length of the **sloping edge** of a pyramid, use **Pythagoras' theorem** to calculate the slant height. You can then calculate the surface area of the pyramid.



Calculate the total surface area of each pyramid, correct to one decimal place.

1 square base: edges 14 cm, sloping edges 18 cm

3 square base: edges 4.2 m, sloping edges 3.8 m

5 rectangular base: 60 cm $\times$ 20 cm, sloping edges 35 cm

7 rectangular base: 5.2 m $\times$ 3.6 m, sloping edges 6.5 m

2 square base: edges 10 cm, sloping edges 15 cm

4 rectangular base: 20 cm $\times$ 12 cm, sloping edges 18 cm

6 rectangular base: 6.5 m $\times$ 8.3 m, sloping edges 10 m

8 rectangular base: 5.8 m $\times$ 4.4 m, sloping edges 7.2 m

A

The total surface area (SA) of a cone is equal to the area of the curved surface plus the area of the circular base.

$$SA = \pi rs + \pi r^2$$

OR
$$= \pi r(s + r)$$

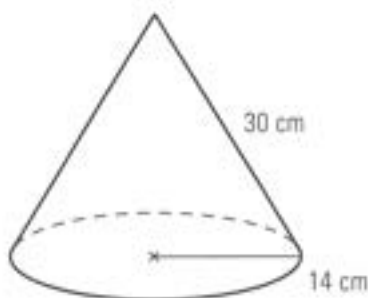
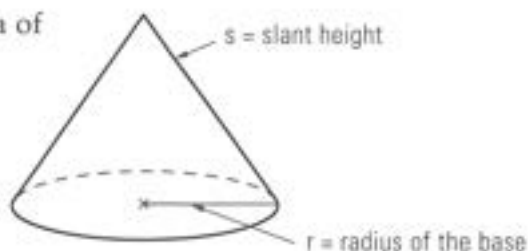
EXAMPLE:

The surface area of this cone

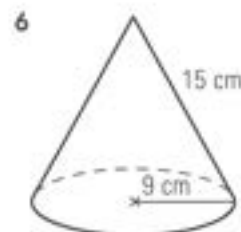
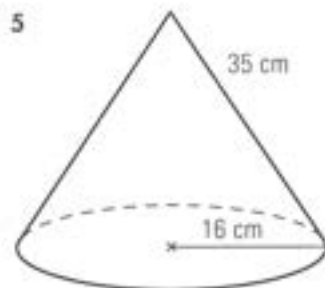
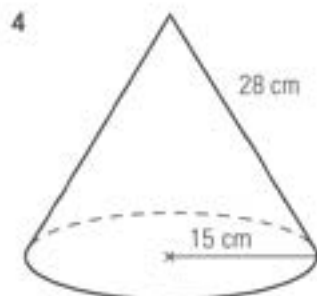
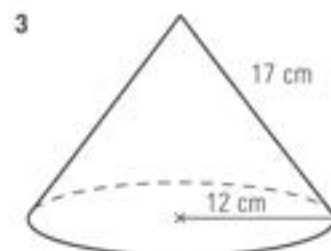
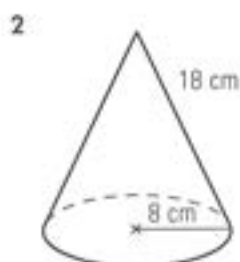
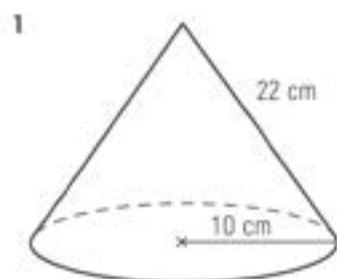
$$= \pi \times 14 \times 30 + \pi \times 14^2$$

$$= 1935.2 \text{ cm}^2$$

(correct to one decimal place)



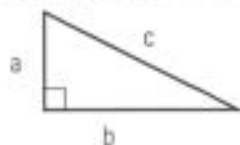
Calculate the total surface area of each cone, correct to one decimal place.



B

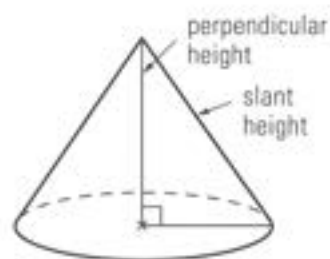
When you are given the perpendicular height of a cone, use **Pythagoras' theorem** to calculate the slant height of the cone. Then calculate the surface area.

Pythagoras' theorem

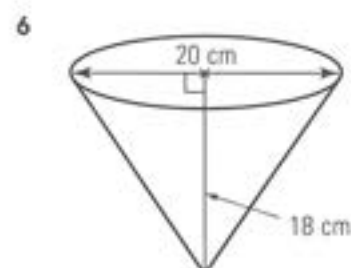
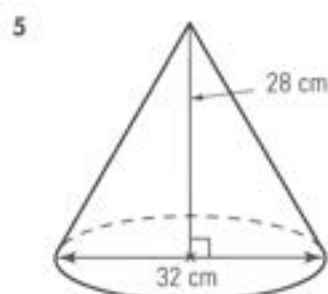
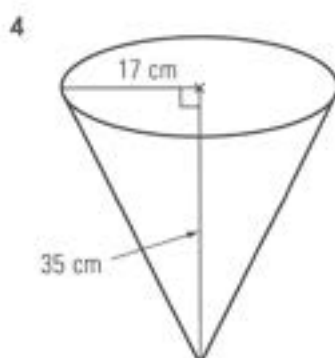
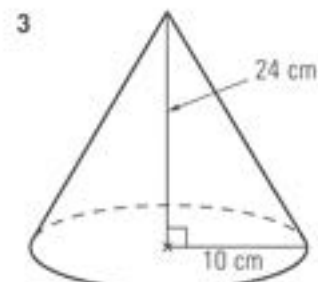
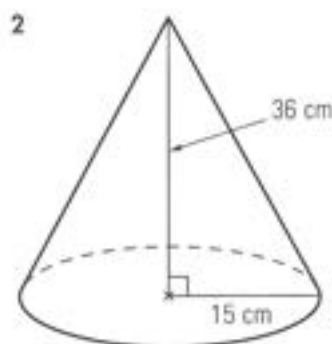
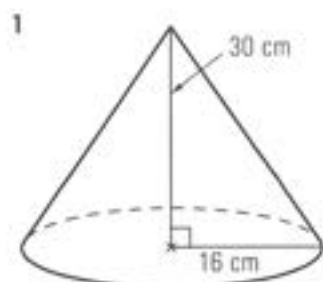


In any right-angled triangle, the square on the hypotenuse is equal to the sum of the squares on the other two sides.

That is: $a^2 + b^2 = c^2$



Calculate the total surface area of each cone, correct to one decimal place.



7 base radius = 60 cm, vertical height = 100 cm

9 base diameter = 24 cm, vertical height = 40 cm

11 base radius = 1.2 m, vertical height = 1.5 m

13 base diameter = 3.4 m, vertical height = 2.3 m

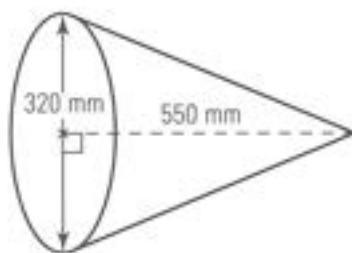
8 base radius = 45 cm, vertical height = 55 cm

10 base diameter = 52 cm, vertical height = 33 cm

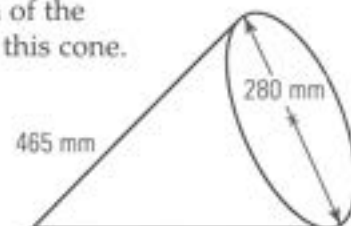
12 base radius = 1.6 m, vertical height = 1.9 m

14 base diameter = 2.9 m, vertical height = 2.5 m

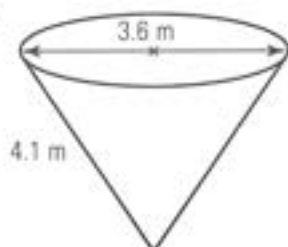
- C** 1 Calculate the area of the curved surface only for a cone with slant height 12 cm and base radius 6.5 cm.
- 3 The nose cone of a rocket has a slant height of 5.3 m and a diameter of 6.4 m. Find the area of the curved surface.
- 5 Calculate the total surface area of this cone.



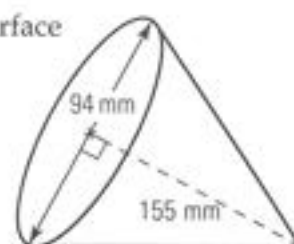
- 2 A conical tent has a base radius of 2.2 m and a slant height of 3.8 m. Find the area of canvas used to make the tent, including the ground sheet.
- 4 Find the total area of a cone with diameter 7.6 m and a perpendicular height of 6.9 m.
- 6 Calculate the area of the curved surface of this cone.



- 7 Calculate the area of the curved surface of this cone.



- 8 Calculate the total surface area of this cone.



A

The surface area of a sphere with radius r units is given by:

$$SA = 4\pi r^2$$

OR
$$= 4 \times \pi \times (\text{radius})^2$$

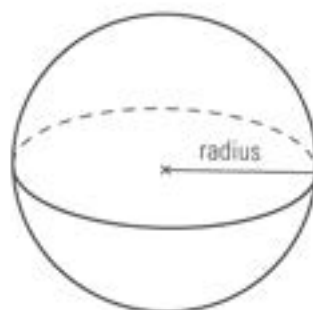
EXAMPLE:

The surface area of a sphere with radius 15 cm

$$SA = 4 \times \pi \times 15^2$$

$$= 2827.4 \text{ cm}^2$$

(correct to one decimal place)



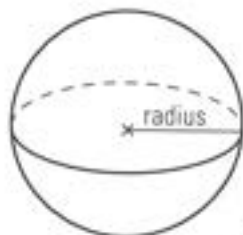
Use the given radius to calculate the surface area of each sphere (correct to one decimal place).

1 radius = 9 cm

2 radius = 12 cm

3 radius = 14 cm

B Calculate the surface area of each sphere (correct to one decimal place).



1 radius = 20 cm

3 radius = 32 cm

5 radius = 1.6 m

7 radius = 0.9 m

9 radius = 85 mm

11 diameter = 26 cm

13 diameter = 25 cm

15 diameter = 1.3 m

17 diameter = 60 mm

2 radius = 25 cm

4 radius = 36 cm

6 radius = 4.5 m

8 radius = 2.8 m

10 radius = 105 mm

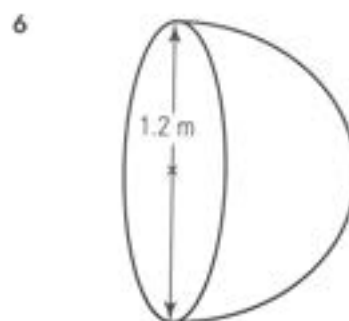
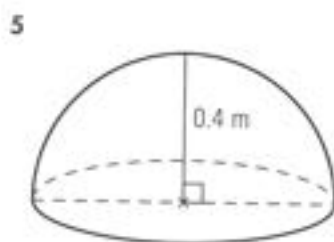
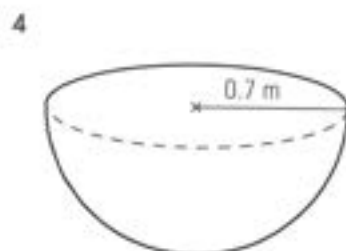
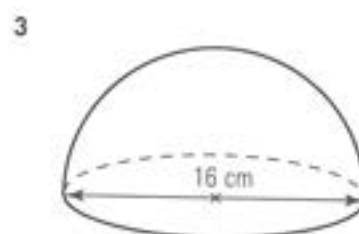
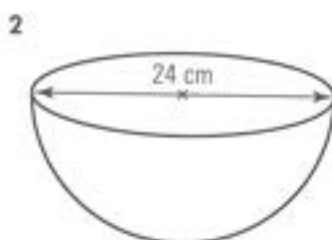
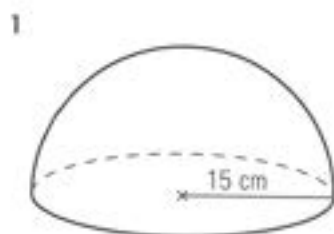
12 diameter = 22 cm

14 diameter = 35 cm

16 diameter = 2.5 m

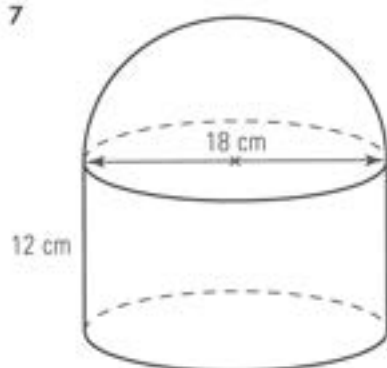
18 diameter = 54 mm

C Calculate the surface area of each hemisphere (half-sphere), correct to one decimal place.
Hint: remember to include the flat surface in your calculations.

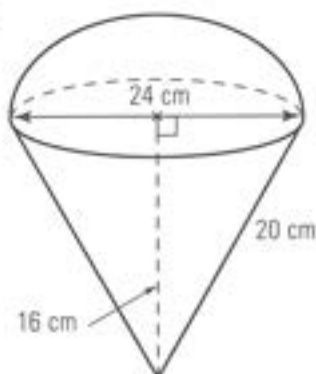


Calculate the total surface area of each solid, correct to one decimal place.

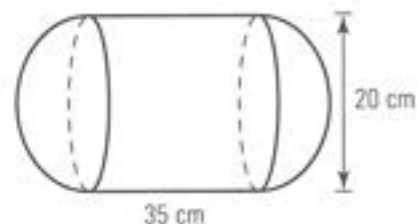
7



8



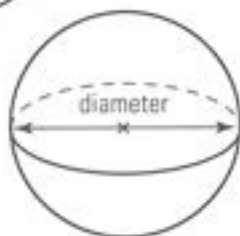
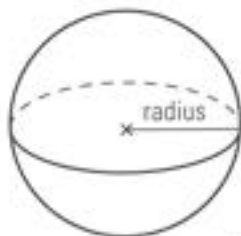
9



- 10 A gas tank consists of a cylinder with identical hemispheres at each end (see Question 9 above). Its overall length is 8 metres and its diameter is 2 metres. Calculate its external surface area, correct to one decimal place.

Riddle: What did the traffic light say to the driver?

Calculate the surface area of each sphere to the nearest square centimetre. Then write the letter beside each question above each correct answer.



A radius = 18 cm

C radius = 10 cm

D radius = 13 cm

K diameter = 24 cm

L diameter = 28 cm

M diameter = 19 cm

G radius = 21 cm

H radius = 27 cm

I radius = 11 m

N diameter = 23 cm

O diameter = 17 cm

T diameter = 37 cm

Note: all answers are cm^2

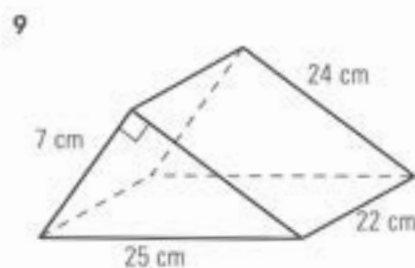
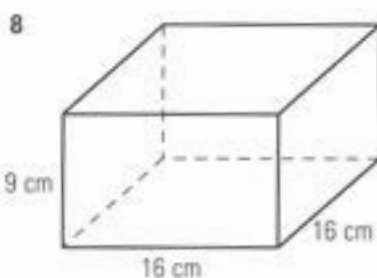
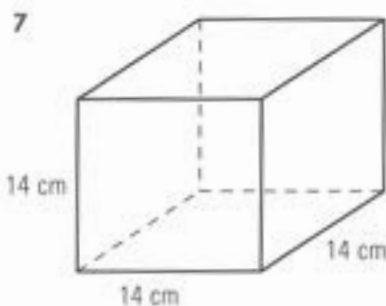
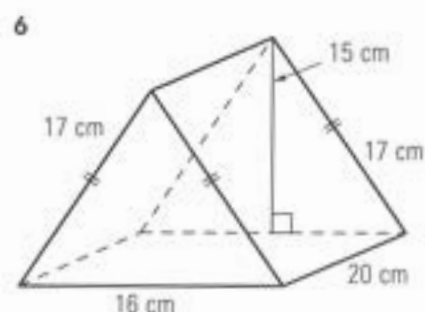
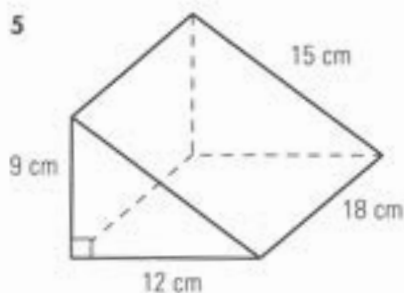
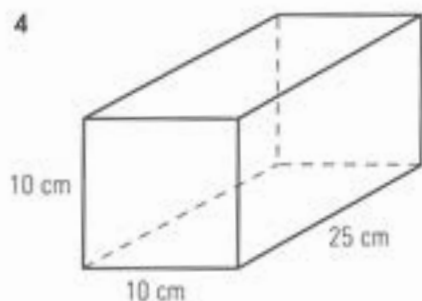
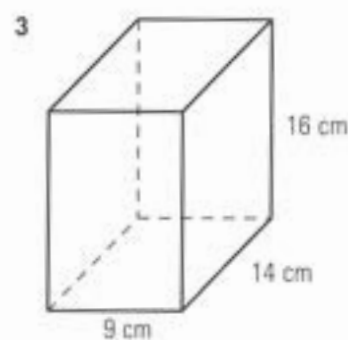
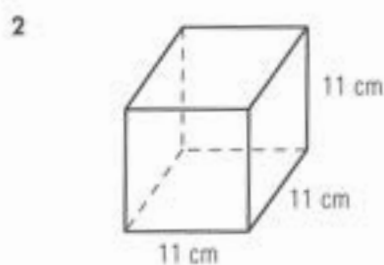
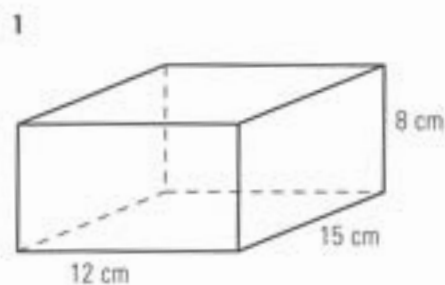
,
2124 908 1662 4301

,
2463 908 908 1810

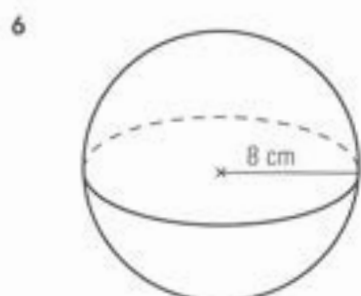
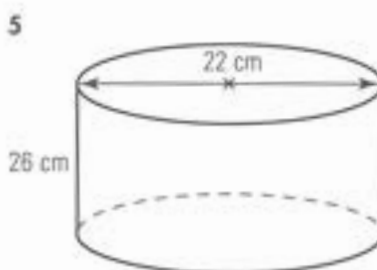
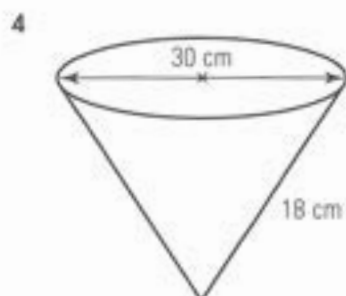
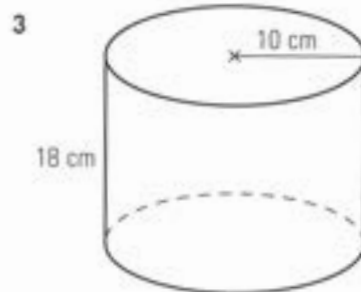
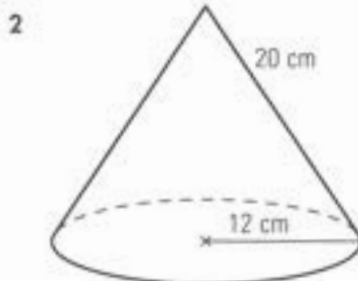
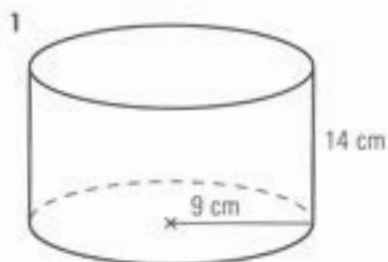
1521 1134

1257 9161 4072 1662 5542 1521 1662 5542

A Calculate the total surface area of each prism.



B Calculate the total surface area of each solid shape, correct to one decimal place.



Calculate the total surface area of each solid shape, correct to one decimal place.

7 rectangular prism: edges 26 cm, 15 cm, 12 cm

9 square prism: edges 14 cm, 14 cm, 18 cm

11 cylinder: diameter 18 cm, height 15 cm

13 cone: base diameter 36 cm, slant height 19 cm

15 sphere: diameter 25 cm

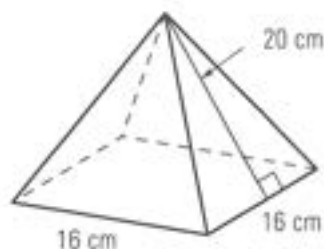
8 cube: edges 20 cm

10 cylinder: radius 10 cm, height 12 cm

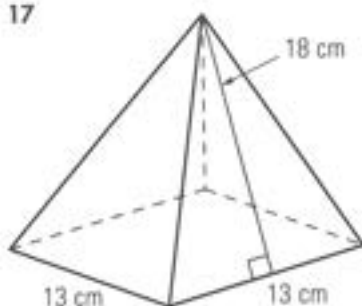
12 cone: base radius 16 cm, slant height 21 cm

14 sphere: radius 17 cm

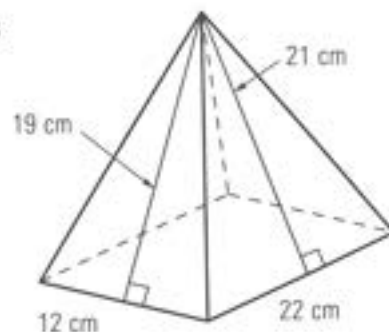
16



17



18



- C** 1 A rectangular room is 5.1 m long, 4.2 m wide and 3.6 m high. Calculate the surface area of the four walls plus the ceiling.
- 2 The surface area of a cube is 384 cm^2 . Calculate the length of each edge of the cube (correct to two decimal places).
- 3 Calculate the area of the curved surface of a cylinder 1.2 m in diameter and 1.8 m high (answer to one decimal place).
- 4 Calculate the area of the curved surface of a cone with base radius 45 cm and slant height 80 cm (to the nearest cm^2).

- 5 Calculate the total surface area of a hemisphere 20 cm in diameter, correct to one decimal place).
- 6 A cylindrical tank, open at the top, is made of sheet metal. If the tank has a diameter of 3.5 m and a height of 3.2 m, find the area of sheet metal that would be used in its construction (answer to two decimal places).
- 7 The inside diameter across the top of a hemispherical bowl is 29 cm. Calculate the surface area of the inside of the bowl, correct to one decimal place.

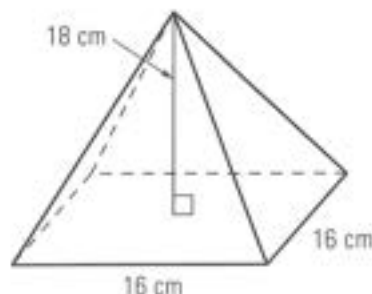
Calculate the total surface area of each solid shape (answer to the nearest square centimetre).

8 rectangular pyramid: base $16 \text{ cm} \times 12 \text{ cm}$, perpendicular height 18 cm

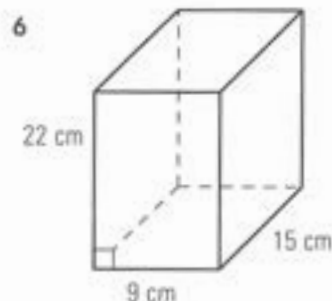
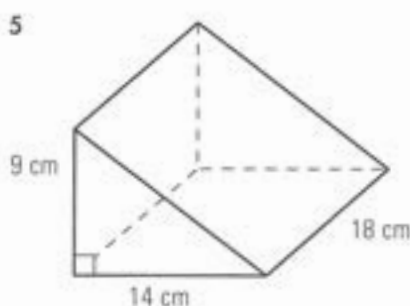
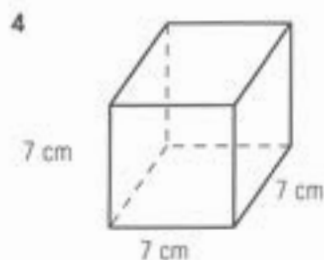
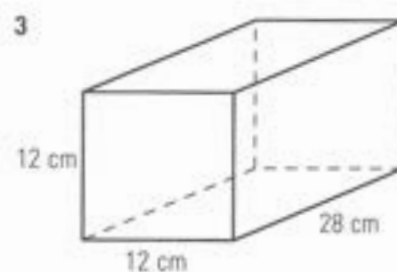
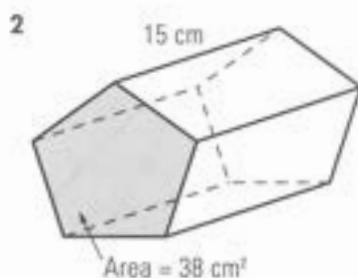
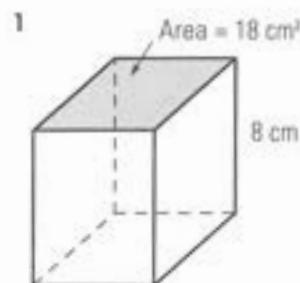
10 cone: base diameter 34 cm, perpendicular height 32 cm



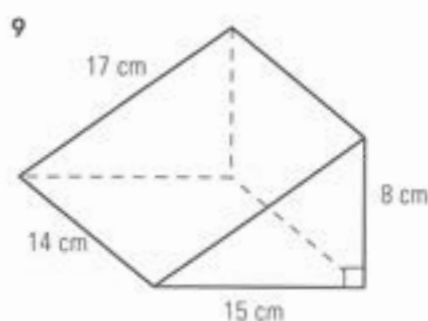
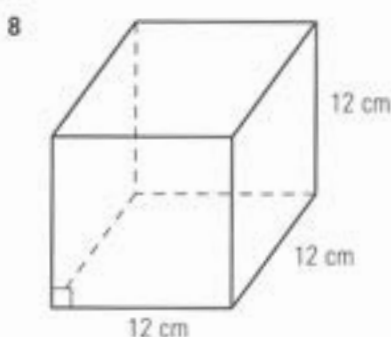
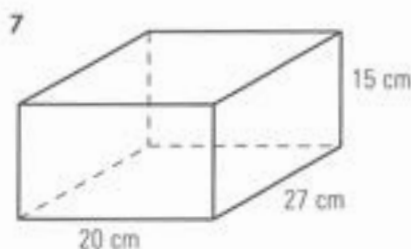
9 square pyramid: base 16 cm, perpendicular height 18 cm



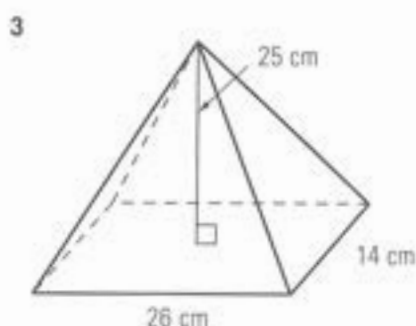
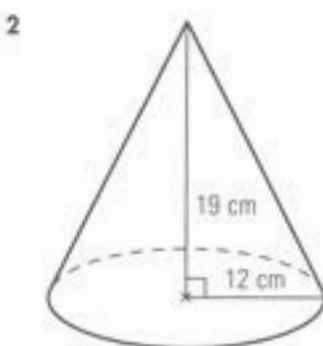
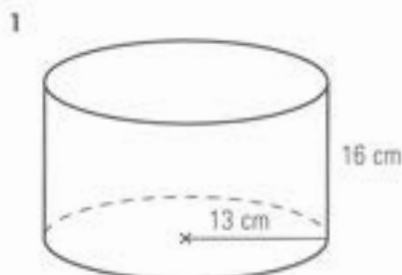
A Calculate the volume of each right prism.



Calculate the total surface area of each right prism.



B Calculate the volume of each solid shape (to one decimal place, where necessary.)



4 rectangular prism: edges 26 cm, 23 cm, 18 cm

6 cylinder: diameter 15 cm, height 16.5 cm

8 sphere: diameter 24 cm

10 square pyramid: base edges 15 cm, perpendicular height 12.5 cm

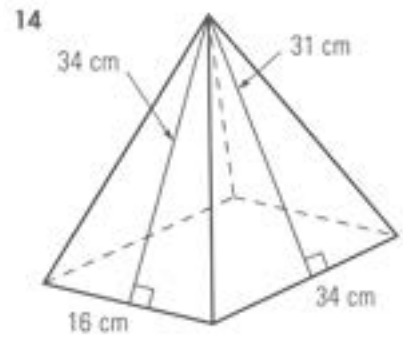
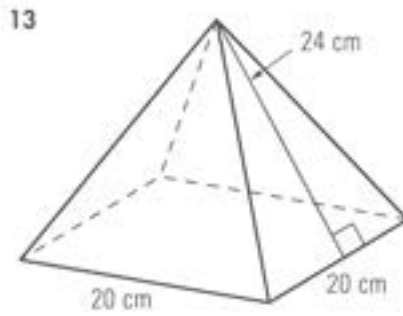
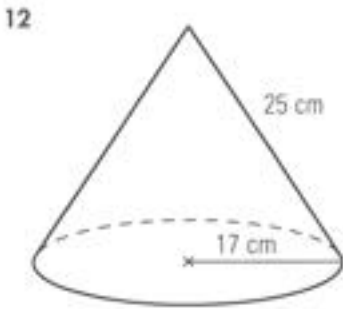
5 cube: edges 3.5 m

7 sphere: radius 14.5 cm

9 cone: base diameter 33 cm, perpendicular height 34 cm

11 hemisphere: radius 15 cm

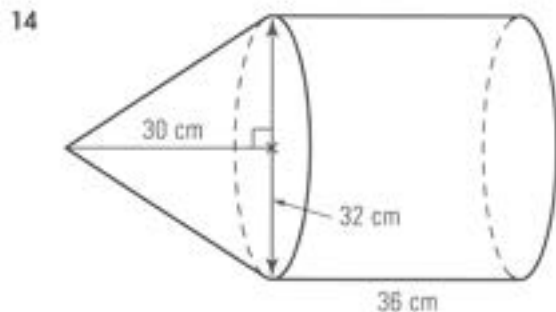
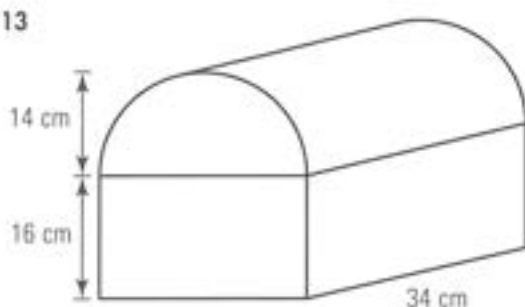
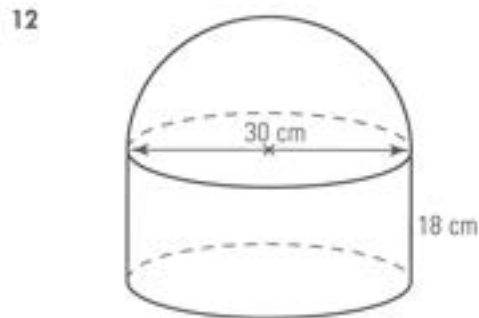
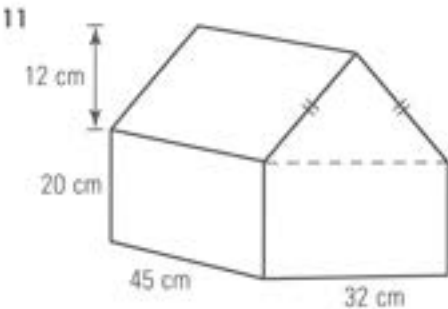
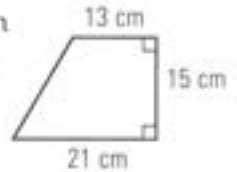
Calculate the total surface area of each solid shape (to one decimal place, where necessary).



- 15 cone: base diameter 26 cm, slant height 19.5 cm 16 sphere: radius 8.5 cm
 17 sphere: diameter 23 cm 18 cylinder: radius 45 cm, height 60 cm
 19 cylinder: diameter 38 cm, height 42 cm 20 hemisphere: diameter 18 cm

C Calculate the volume and the total surface area of each solid shape (answer to two decimal places, where necessary).

- 1 rectangular prism: edges 21 cm, 11.5 cm, 9.5 cm 2 cube: edges 15.5 cm
 3 cylinder: diameter 14.4 cm, height 18.5 cm 4 sphere: diameter 21.6 cm
 5 cone: base radius 34 cm, slant height 45 cm 6 cone: base radius 29 cm, perpendicular height 35 cm
 7 A prism 40 cm long in which the cross-section is an isosceles triangle where the equal sides measure 25 cm and the third side 20 cm.
 8 A square pyramid with base edges 36 cm and perpendicular height 24 cm
 9 A square pyramid with base edges measuring 32 cm and slant height 34 cm
 10 A trapezoidal prism 45 cm long with a cross-section as shown



Riddle: Where do cows go for their holidays?

Calculate the surface area of each solid shape.
Then write the letter beside each solid above each correct answer.
(Answer to the nearest square metre.)

- K** sphere: radius 2.5 m

M cylinder: diameter 3.1 m, height 3.5 m

O cube: edges 2.5 m

R cone: base diameter 2.8 m, slant height 3 m

Y rectangular prism: edges 2.8 m, 1.8 m, 1.5 m

49 m ²	38 m ²	38 m ²

24 m ²	38 m ²	19 m ²	79 m ²

Riddle: Why did the student take his maths homework to the movies?

Calculate the volume of each solid shape to the nearest cubic metre.
Then write the letter beside each question above each correct answer.
(Hint: change all units to metres **before** you multiply.)

- A** rectangular prism: edges 4.2 m, 2.9 m, 2.3 m

B cube: edges 210 cm

C triangular prism: area of cross-section = 4.5 m², height = 2.6 m

D square prism: edges 140 cm, 140 cm, 2.6 m

E cylinder: radius = 170 cm, height = 3.9 m

H cylinder: diameter = 3.1 m, height 2.7 m

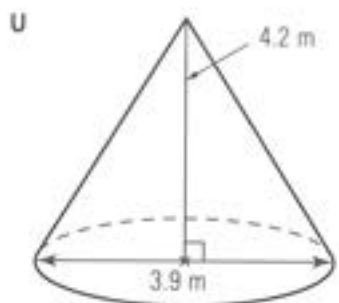
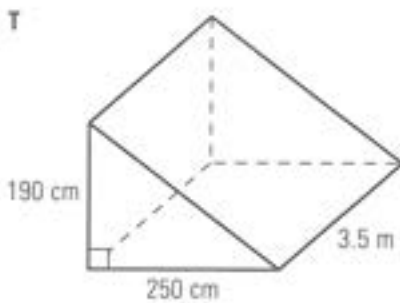
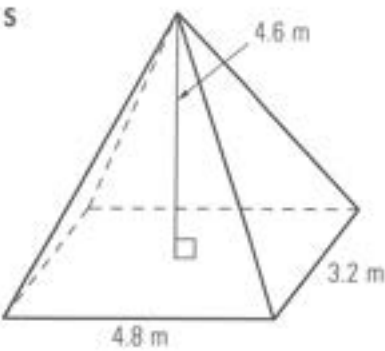
I hexagonal prism: area of cross-section = 5.3 m², height = 1.9 m

L square pyramid: base edge = 180 cm, height 2.5 m

M cone: radius = 2.2 m, perpendicular height = 2.7 m

O sphere: radius = 3.8 m

R sphere: diameter = 2.4 m



9	35	12	28

20	10	24

8	35	28	12

8	230	3	5

20	10	14

8	230

5	28	8	35

10	8

Note: all answers in m³

A

Length

10 millimetres (10 mm) = 1 centimetre (1 cm)
 100 centimetres (100 cm) = 1 metre (1 m)
 1000 millimetres (1000 mm) = 1 metre (1 m)
 1000 metres (1000 m) = 1 kilometre (1 km)

Change each measurement to the given units.

- | | | | |
|--------------------|-------------------|--------------------|--------------------|
| 1 150 mm = __ cm | 2 230 mm = __ cm | 3 45 mm = __ cm | 4 700 cm = __ m |
| 5 3 400 cm = __ m | 6 540 cm = __ m | 7 80 cm = __ m | 8 2 000 mm = __ m |
| 9 14 500 mm = __ m | 10 750 mm = __ m | 11 5 000 m = __ km | 12 4 060 m = __ km |
| 13 12 cm = __ mm | 14 30 cm = __ mm | 15 6.35 m = __ cm | 16 14.7 m = __ cm |
| 17 3 km = __ m | 18 8.4 km = __ m | 19 0.65 km = __ m | 20 9 m = __ mm |
| 21 3.56 m = __ mm | 22 0.35 m = __ mm | 23 0.8 cm = __ mm | 24 184 cm = __ mm |

B

Area

100 mm² = 1 cm²
 10 000 cm² = 1 m²
 1 000 000 mm² = 1 m²
 10 000 m² = 1 hectare (1 ha)

Change each measurement to the given units.

- | | | | |
|--|--|---|--|
| 1 30 000 cm ² = __ m ² | 2 5 600 cm ² = __ m ² | 3 800 cm ² = __ m ² | 4 67 000 cm ² = __ m ² |
| 5 40 000 m ² = __ ha | 6 5 000 m ² = __ ha | 7 950 m ² = __ ha | 8 7 m ² = __ cm ² |
| 9 6.2 m ² = __ cm ² | 10 0.3 m ² = __ cm ² | 11 1.25 m ² = __ cm ² | 12 4 ha = __ m ² |
| 13 2.5 ha = __ m ² | 14 600 mm ² = __ cm ² | 15 5 000 mm ² = __ cm ² | 16 860 mm ² = __ cm ² |
| 17 180 000 mm ² = __ m ² | 18 3 000 000 mm ² = __ m ² | 19 58 cm ² = __ mm ² | 20 14 cm ² = __ mm ² |
| 21 9.3 cm ² = __ mm ² | 22 6 m ² = __ mm ² | 23 4.2 m ² = __ mm ² | 24 0.25 m ² = __ mm ² |

C

Volume

1 000 000 cm³ = 1 m³
 1 000 mm³ = 1 cm³

Change each measurement to the given units.

- | | | | |
|--|--|---|--|
| 1 2 cm ³ = __ mm ³ | 2 11 cm ³ = __ mm ³ | 3 5.2 cm ³ = __ mm ³ | 4 1.6 cm ³ = __ mm ³ |
| 5 10.8 cm ³ = __ mm ³ | 6 3 m ³ = __ cm ³ | 7 1.5 m ³ = __ cm ³ | 8 4.01 m ³ = __ cm ³ |
| 9 0.75 m ³ = __ cm ³ | 10 2.43 m ³ = __ cm ³ | 11 4 000 mm ³ = __ cm ³ | 12 3 500 mm ³ = __ cm ³ |
| 13 12 000 mm ³ = __ cm ³ | 14 2 850 mm ³ = __ cm ³ | 15 315 000 mm ³ = __ cm ³ | 16 6 000 000 cm ³ = __ m ³ |
| 17 2 700 000 cm ³ = __ m ³ | 18 900 000 cm ³ = __ m ³ | 19 320 000 cm ³ = __ m ³ | 20 7 400 000 cm ³ = __ m ³ |

1 Volume of a right prism

- A 1 380 cm³ 2 216 cm³ 3 550 cm³ 4 630 cm³
5 560 cm³ 6 972 cm³
- B 1 2688 cm³ 2 8250 cm³ 3 5616 cm³ 4 1728 cm³
5 2178 cm³ 6 3458 cm³ 7 1440 cm³ 8 3630 cm³
9 5600 cm³ 10 1500 cm³ 11 3456 cm³ 12 3200 cm³
- C 1 12 cm 2 11 cm 3 15 cm 4 9 cm 5 16 cm
6 14 cm 7 12 cm 8 1.8 m 9 2.5 m 10 3.2 m
11 7.36 m³ 12 7.02 m³ 13 9.12 m³ 14 11.13 m³
15 33.25 m³ 16 15.3 m³ 17 5.55 m³ 18 12.985 m³
19 21.96 m³

2 Volume of a rectangular prism

- A 1 1536 cm³ 2 3360 cm³ 3 1080 cm³ 4 9240 cm³
5 13 440 cm³ 6 5600 cm³
- B 1 3024 cm³ 2 2970 cm³ 3 1232 cm³ 4 4680 cm³
5 2464 cm³ 6 4576 cm³ 7 6600 cm³ 8 3024 cm³
9 127 500 mm³ 10 77 000 mm³ 11 130 000 mm³
12 794 750 mm³ 13 15.75 m³ 14 17.664 m³
15 17.732 m³ 16 16.416 m³ 17 16.2 m³ 18 23.4 m³
19 5.88 m³ 20 24.219 m³
- C 1 8 cm 2 9 cm 3 24 cm 4 30 cm 5 25 cm 6 12 cm
7 4 m 8 2 m 9 1.5 m 10 1.6 m 11 3.78 m³
12 1.6 m³ 13 11.648 m³ 14 14.742 m³ 15 28.08 m³
16 15.048 m³ 17 44.2 m³ 18 16.56 m³ 19 42 m³
20 4.45 m³ 21 3.213 m³ 22 8.372 m³

3 Volume of a triangular prism

- A 1 216 cm³ 2 448 cm³ 3 1230 cm³ 4 2720 cm³
5 3600 cm³ 6 3190 cm³
- B 1 2400 cm³ 2 360 cm³ 3 4500 cm³ 4 1152 cm³
5 1200 cm³ 6 2688 cm³ 7 2640 cm³ 8 6720 cm³
9 2700 cm³ 10 2640 cm³ 11 2737 cm³ 12 3675 cm³
- C 1 19 200 cm³ 2 22 680 cm³ 3 7296 cm³ 4 4536 cm³
5 9600 cm³ 6 16 200 cm³

4 Volume of a cylinder

- A 1 8143 cm³ 2 3817 cm³ 3 9852 cm³ 4 4021 cm³
5 19 302 cm³ 6 8482 cm³
- B 1 37 699.1 cm³ 2 57 780.2 cm³ 3 29 487.0 cm³
4 25 421.8 cm³ 5 50 893.8 cm³ 6 15 095.4 cm³
7 9533.2 cm³ 8 44 257.2 cm³ 9 32 501.3 cm³
10 30.6 m³ 11 2.7 m³ 12 4 539 601.4 mm³
13 1 858 252.1 mm³ 14 16.3 m³ 15 5.9 m³
16 2 251 343.8 mm³ 17 2 385 646.9 mm³
- C 1 1.6 m³ 2 6 379 396.6 mm³ 3 424 115.0 cm³
4 188 495.6 cm³ 5 8 cm 6 785.4 cm³, no 7 95 cm
8 15.9 cm

5 Review: volumes of prisms

- A 1 432 cm³ 2 490 cm³ 3 336 cm³ 4 460 cm³
5 720 cm³ 6 480 cm³
- B 1 3240 cm³ 2 4913 cm³ 3 3200 cm³ 4 4680 cm³
5 6720 cm³ 6 10 010 cm³ 7 12 666.9 cm³
8 27 344.4 cm³ 9 2924.8 cm³ 10 12 768 cm³
11 2197 cm³ 12 6400 cm³ 13 10 752 cm³
14 6842.4 cm³ 15 20 106.2 cm³ 16 4.4 m³ 17 12.8 m³
18 6.9 m³ 19 14.2 m³ 20 3.2 m³ 21 3840 cm³
22 8208 cm³ 23 12 300 cm³ 24 1767.1 cm³
25 4704.7 cm³
- C I've got you covered. C 2145 cm³ D 1134 cm³
E 1000 cm³ G 2592 cm³ I 4080 cm³ O 3375 cm³
R 4072 cm³ T 3181 cm³ U 1440 cm³ V 3024 cm³
Y 1560 cm³

6 Volume of a pyramid

- A 1 2400 cm³ 2 840 cm³ 3 1600 cm³ 4 1800 cm³
5 378 cm³ 6 528 cm³ 7 750 cm³ 8 605 cm³
- B 1 2100 cm³ 2 672 cm³ 3 533.3 cm³ 4 330 cm³
5 597.3 cm³ 6 1437.3 cm³ 7 161.5 m³ 8 9.6 m³
9 28.3 m³ 10 5.2 m³ 11 1280 cm³ 12 3072 cm³
13 3200 cm³ 14 1296 cm³ 15 10 240 cm³
- C 1 3505.6 cm³ 2 4916.1 cm³ 3 3358.6 cm³ 4 360 cm³
5 866.0 cm³ 6 15 cm 7 9 m 8 323.2 cm³
9 982.2 cm³ 10 13.9 m³ 11 1096.9 cm³ 12 54.2 m³

7 Volume of a cone

- A 1 1885.0 cm³ 2 1357.2 cm³ 3 2563.5 cm³
4 5183.6 cm³ 5 6702.1 cm³ 6 938.3 cm³
- B 1 4926.0 cm³ 2 4222.3 cm³ 3 1675.5 cm³
4 2412.7 cm³ 5 4241.2 cm³ 6 5131.3 cm³
7 4712.4 cm³ 8 4105.0 cm³ 9 6126.1 cm³
10 9382.9 cm³ 11 2407.5 cm³ 12 7263.4 cm³
13 4247.4 cm³ 14 9160.9 cm³
- C 1 8043 cm³ 2 2513 cm³ 3 8482 cm³ 4 2959 cm³
5 8345 cm³ 6 6803 cm³ 7 8560 cm³ 8 9577 cm³
9 11 739 cm³ 10 10 818 cm³

8 Volume of a sphere

- A 1 4188.8 cm³ 2 3053.6 cm³ 3 14 137.2 cm³
4 7238.2 cm³ 5 904.8 cm³ 6 11 494.0 cm³
- B 1 24 429.0 cm³ 2 65 449.8 cm³ 3 17.2 m³ 4 179.6 m³
5 3.1 m³ 6 51.0 m³ 7 1 150 346.5 mm³
8 775 734.6 mm³ 9 2144.7 cm³ 10 5575.3 cm³
11 9202.8 cm³ 12 8181.2 cm³ 13 22 449.3 mm³
14 0.9 m³ 15 2.1 m³ 16 8.2 m³ 17 113 097.3 mm³
18 212 174.8 mm³

- C** 1 10 289.76 cm³ 2 19 396.19 cm³ 3 8578.64 cm³
 4 2.78 m³ 5 10.28 m³ 6 0.19 m³ 7 3004.1 cm³
 8 8999.3 cm³ 9 25 cm 10 15.8 m³ 11 7.8 cm
 12 7.3 cm 13 7.6 cm 14 8.3 cm 15 8.5 cm
 16 9.2 cm 17 1.3 m 18 1.4 m 19 1.5 m 20 16.3 m
 21 15.5 cm 22 17.4 cm 23 2.7 m 24 2.6 m 25 3.0 m

9 Review: volume

- A** 1 5280 cm³ 2 6615 cm³ 3 464 cm³ 4 414 cm³
 5 2880 cm³ 6 3375 cm³ 7 2464 cm³ 8 8580 cm³
 9 5832 cm³ 10 3332 cm³
- B** 1 1080 cm³ 2 2520 cm³ 3 972 cm³ 4 3769.9 cm³
 5 7634.1 cm³ 6 7238.2 cm³ 7 2640 cm³ 8 17 640 cm³
 9 2160 cm³ 10 2800 cm³ 11 2408.6 cm³
 12 5336.5 cm³ 13 4326.0 cm³ 14 23 561.9 cm³
 15 672 cm³ 16 5184 cm³ 17 4825.5 cm³
 18 2144.7 cm³ 19 6370.6 cm³ 20 10 306.0 cm³
- C** 1 11 423.0 cm³ 2 6120 cm³ 3 21 756.6 cm³
 4 988 litres 5 A, 1319.5 cm³ 6 18 818.8 cm³
 7 12 m 8 9 m 9 50 cm 10 8042.5 cm³
 11 25 132.7 cm³ 12 672 cm³ **Riddle** The pork chop
 C 2898 E 7000 H 9047.8 K 2016 O 3048.6 P 7238.2
 R 2261.9 T 2144.7

10 Surface area of a rectangular prism

- A** 1 708 cm² 2 1048 cm² 3 2510 cm² 4 1392 cm²
- B** 1 832 cm² 2 1576 cm² 3 654 cm² 4 2732 cm²
 5 3536 cm² 6 1948 cm² 7 16 600 mm² 8 11 050 mm²
 9 24 700 mm²
- C** 1 1272 cm² 2 1054 cm² 3 1844 cm² 4 2056 cm²
 5 13 450 mm² 6 21 500 mm² 7 26 050 mm²
 8 39.1 m² 9 79.84 m² 10 57.6 m² 11 34.98 m²
 12 67.54 m²

11 Surface area of a triangular prism

- A** 1 468 cm² 2 1440 cm² 3 408 cm² 4 2100 cm²
- B** 1 840 cm² 2 810 cm² 3 1536 cm² 4 984 cm²
 5 1340 cm² 6 2496 cm² 7 1416 cm²
- C** 1 4960 cm² 2 5616 cm² 3 3120 cm² 4 2064 cm²
 5 3480 cm² 6 5220 cm²

12 Surface areas of various prisms

- A** 1 1208 cm² 2 2078 cm² 3 1692 cm² 4 1366 cm²
 5 2412 cm² 6 2056 cm²
- B** 1 1332 cm² 2 1734 cm² 3 1480 cm² 4 1928 cm²
 5 2142 cm² 6 1944 cm² 7 2952 cm² 8 2720 cm²
 9 6480 cm² 10 3320 cm² 11 1350 cm² 12 2112 cm²
 13 536 cm² 14 2450 cm² 15 3880 cm² 16 1820 cm²

- 17 1448 cm² 18 3220 cm² 19 16.06 m² 20 44.08 m²
 21 37.5 m² 22 49.92 m² 23 21.9 m² 24 47.88 m²

C Because he wanted to become a smartie.

- A** 1472 cm² **B** 2760 cm² **C** 2750 cm² **D** 5400 cm²
E 3592 cm² **H** 3194 cm² **I** 1950 cm² **M** 3200 cm²
N 1350 cm² **O** 3750 cm² **R** 3108 cm² **S** 1568 cm²
T 2380 cm² **U** 3936 cm² **W** 4960 cm²

13 Surface area of a cylinder

- A** 1 2261.9 cm² 2 1357.2 cm² 3 2638.9 cm²
 4 1407.4 cm² 5 4021.2 cm² 6 2544.7 cm²
- B** 1 6283.2 cm² 2 8293.8 cm² 3 8953.5 cm²
 4 7389.0 cm² 5 7514.7 cm² 6 5969.0 cm²
 7 4806.6 cm² 8 7690.6 cm² 9 3457.3 cm²
 10 2697.1 cm² 11 6982.2 cm² 12 5650.2 cm²
 13 49.8 m² 14 13.0 m² 15 35.5 m² 16 178 913.7 mm²
 17 306 776.5 mm² 18 68 964.2 mm² 19 78.0 m²
 20 164.4 m² 21 18.6 m² 22 83 291.5 mm²
 23 80 778.2 mm² 24 121 579.6 mm²
- C** 1 2827.4 cm² 2 4750.1 cm² 3 1696.5 cm² 4 4775.2 cm²
 5 11 231.2 cm² 6 2563.5 cm² 7 4222.3 cm²
 8 989.6 cm² 9 717.1 cm² 10 3317.5 cm²
 11 2136.3 cm² 12 3901.9 cm² 13 3317.5 cm²
 14 140 743.4 mm² 15 114 707.4 mm² 16 10 m² 17 5 m²

14 Surface area of a pyramid

- A** 1 705 cm² 2 1120 cm² 3 1080 cm² 4 1825 cm²
- B** 1 1536 cm² 2 3200 cm² 3 1440 cm² 4 864 cm²
 5 1606 cm² 6 2048 cm² 7 1108.8 cm² 8 1568.8 cm²
 9 665.3 cm² 10 198.1 m²
- C** 1 660.2 cm² 2 382.8 cm² 3 44.3 m² 4 743.0 cm²
 5 2952.6 cm² 6 191.0 m² 7 72.2 m² 8 93.9 m²

15 Surface area of a cone

- A** 1 1005.3 cm² 2 653.5 cm² 3 1093.3 cm²
 4 2026.3 cm² 5 2563.5 cm² 6 678.6 cm²
- B** 1 2513.3 cm² 2 2544.7 cm² 3 1131.0 cm²
 4 2986.0 cm² 5 2425.3 cm² 6 961.1 cm²
 7 34 928.2 cm² 8 16 408.1 cm² 9 2026.8 cm²
 10 5555.3 cm² 11 11.8 m² 12 20.5 m² 13 24.4 m²
 14 19.8 m²
- C** 1 245.0 cm² 2 41.5 m² 3 53.3 m² 4 139.4 m²
 5 368 345.5 mm² 6 266 092.9 mm² 7 72.2 m²
 8 93.9 m²

16 Surface area of a sphere

- A** 1 1017.9 cm² 2 1809.6 cm² 3 2463.0 cm²

- B** 1 5026.5 cm³ 2 7854.0 cm³ 3 12 868.0 cm³
 4 16 286.0 cm³ 5 32.2 m³ 6 254.5 m³ 7 10.2 m³
 8 98.5 m³ 9 90 792.0 mm³ 10 138 544.2 mm³
 11 2123.7 cm³ 12 1520.5 cm³ 13 1963.5 cm³
 14 3848.5 cm³ 15 5.3 m³ 16 19.6 m³
 17 11 309.7 mm³ 18 9160.9 mm³
- C** 1 2120.6 cm³ 2 1357.2 cm³ 3 603.2 cm³
 4 4.6 m³ 5 1.5 m³ 6 3.4 m³ 7 1442.0 m³
 8 1658.8 cm³ 9 3455.8 cm³ 10 50.3 m³ Riddle Don't
 look, I'm changing. A 4072 C 1257 D 2124 G 5542
 H 9161 I 1521 K 1810 L 2463 M 1134 N 1662
 O 908 T 4301

17 Review: surface areas of 3D shapes

- A** 1 792 cm² 2 726 cm² 3 988 cm² 4 1200 cm²
 5 756 cm² 6 1240 cm² 7 1176 cm² 8 1088 cm²
 9 1400 cm²
- B** 1 1300.6 cm² 2 1206.4 cm² 3 1759.3 cm²
 4 1555.1 cm² 5 2557.3 cm² 6 804.2 cm² 7 1764 cm²
 8 2400 cm² 9 1400 cm² 10 1382.3 cm²
 11 1357.2 cm² 12 1859.8 cm² 13 2092.3 cm²
 14 3631.7 cm² 15 1963.5 cm² 16 896 cm²
 17 637 cm² 18 954 cm²
- C** 1 88.38 m² 2 8 cm 3 6.8 m² 4 11 310 cm²
 5 942.5 cm² 6 44.81 m² 7 1321.0 cm² 8 886.4 cm²
 9 731.9 cm² 10 2843.4 cm²

18 Review: volumes and surface areas

- A** 1 144 cm³ 2 570 cm³ 3 4032 cm³ 4 343 cm³
 5 1134 cm³ 6 2970 cm³ 7 2490 cm³ 8 864 cm³
 9 680 cm³
- B** 1 8494.9 cm³ 2 2865.1 cm³ 3 3033.3 cm³
 4 10 764 cm³ 5 42.9 m³ 6 2915.8 cm³
 7 12 770.1 cm³ 8 7238.2 cm³ 9 9693.4 cm³
 10 937.5 cm³ 11 7068.6 cm³ 12 2243.1 cm³
 13 1360 cm³ 14 2142 cm³ 15 1327.3 cm³
 16 907.9 cm³ 17 1661.9 cm³ 18 29 688.1 cm³
 19 7282.2 cm³ 20 763.4 cm³
- C** 1 V = 2294.25 cm³, SA = 1100.5 cm²
 2 V = 3723.88 cm³, SA = 1441.5 cm²
 3 V = 3012.91 cm³, SA = 1162.64 cm²
 4 V = 5276.67 cm³, SA = 1465.74 cm²
 5 V = 35 685.87 cm³, SA = 8438.32 cm²
 6 V = 30 824.26 cm³, SA = 6783.15 cm²
 7 V = 9165.15 cm³, SA = 3258.26 cm²
 8 V = 10 368 cm³, SA = 3456 cm²
 9 V = 10 240 cm³, SA = 3200 cm²
 10 V = 11 475 cm³, SA = 3480 cm²
 11 V = 37 440 cm³, SA = 6704 cm²
 12 V = 19 792.03 cm³, SA = 3817.04 cm²
 13 V = 25 699.79 cm³, SA = 5047.15 cm²

$$14 V = 36\,995.40 \text{ cm}^3, SA = 6132.39 \text{ cm}^2$$

Riddle Moo York K 79 m³ M 49 m³ O 28 m³ R 19 m³
 Y 24 m³ Riddle Because his teacher told him to date
 it. A 28 m³ B 9 m³ C 12 m³ D 5 m³ E 35 m³ H 20
 m³ I 10 m³ L 3 m³ M 14 m³ O 230 m³ R 7 m³ S
 24 m³ T 8 m³ U 17 m³

19 Equivalent units

- A** 1 15 2 23 3 4.5 4 7 5 34 6 5.4 7 0.8 8 2 9 14.5
 10 0.75 11 5 12 4.06 13 120 14 300 15 635
 16 1 470 17 3 000 18 8 400 19 650 20 9 000
 21 3 560 22 350 23 8 24 1 840
- B** 1 3 2 0.56 3 0.08 4 6.7 5 4 6 0.5 7 0.095
 8 70 000 9 62 000 10 3 000 11 12 500 12 40 000
 13 25 000 14 6 15 50 16 8.6 17 0.18 18 3
 19 5 800 20 1 400 21 930 22 6 000 000
 23 4 200 000 24 250 000
- C** 1 2 000 2 11 000 3 5 200 4 1 600 5 10 800
 6 3 000 000 7 1 500 000 8 4 010 000 9 750 000
 10 2 430 000 11 4 12 3.5 13 12 14 2.85 15 315
 16 6 17 2.7 18 0.9 19 0.32 20 7.4

Formulae

Volume

Volume of a prism = Area $\times$ perpendicular height

$$V = A h$$

Volume of a rectangular prism = length $\times$ breadth $\times$ height

$$V = l b h$$

Volume of a cylinder = $\pi \times (\text{radius})^2 \times \text{height}$

$$V = \pi r^2 h$$

Volume of a pyramid = $\frac{1}{3} \times \text{Area of base} \times \text{height}$

$$V = \frac{1}{3} A h$$

Volume of a cone = $\frac{1}{3} \times \pi \times (\text{radius})^2 \times \text{height}$

$$V = \frac{1}{3} \pi r^2 h$$

Volume of a sphere = $\frac{4}{3} \times \pi \times (\text{radius})^3$

$$V = \frac{4}{3} \pi r^3$$

Surface area

The surface area of a solid shape = the sum of the areas of its faces

The surface area of a rectangular prism = the sum of the areas of its faces

$$SA = 2(lb + lh + bh)$$

Total surface area of a cylinder = area of the curved surface + area of two circular ends

$$SA = 2\pi r h + 2\pi r^2$$

Total surface area of a cone = area of the curved surface + area of the circular base

$$SA = \pi r s + \pi r^2$$

Surface area of a sphere = area of the curved surface

$$SA = 4\pi r^2$$

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Volume of a pyramid, cylinder and cone
Volume of a sphere and hemisphere

Surface areas of various prisms including rectangular and triangular
Surface area of a pyramid, cylinder and cone
Surface area of a sphere and hemisphere

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